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LEONARDO TADEU LIMA FRANCO

Deep Learning-Based Reconstruction of Railway Track Irregularities from Acceleration Data Considering Velocity and Mass Effects

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Leonardo Tadeu Lima Franco

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Orientador: Auteliano Antunes dos Santos

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Resumo

Esta tese propõe uma abordagem de deep learning para a reconstrução de irregularidades da via a partir de dados de aceleração. Esse trabalho investiga, primeiramente, a influência da velocidade e da massa do vagão nas medições de aceleração, com o objetivo de remover o efeito dessas condições de operação para aumentar a robustez do método. A partir desse estudo, um modelo de correção foi criado para corrigir dados de medição de vagões instrumentados. De posse dos dados corrigidos, um modelo de deep learning foi treinado com os dados corrigidos do vagão instrumentado da simulação para obter as irregularidades geométricas da via. Este modelo depois foi avaliado em um conjunto de dados reais de vagão instrumentado e irregularidades geométricas. Com as irregularidades reconstruídas, é possível identificar locais onde os limites de segurança geométricas foram ultrapassadas com base nos dados de aceleração. Consequentemente, os dados medidos pelo vagão instrumentado podem ser usados para identificar defeitos geométricos.

Palavras-chave: monitoramento de vias ferroviárias; irregularidades geométricas da via; deep learning; machine learning; simulação de multicorpos; efeito da massa e velocidade;

Abstract

This thesis proposes a deep learning approach for the reconstruction of track irregularities from acceleration data. The study first investigates the influence of train speed and wagon mass on acceleration measurements, with the goal of removing the effect of these operational conditions to increase the robustness of the method. Based on this study, a correction model was developed to adjust measurement data from instrumented wagons. With the corrected data, a deep learning model was trained using the corrected simulation data from the instrumented wagon to obtain the geometric track irregularities. This model was then evaluated on a real dataset of instrumented wagon measurements and geometric irregularities. With the reconstructed irregularities, it becomes possible to identify locations where the geometric safety limits have been exceeded based on acceleration data. Consequently, the data measured by the instrumented wagon can be used to identify geometric defects.

Keywords: railway track monitoring; track geometric irregularities; deep learning; machine learning; multibody simulation; effect of mass and speed.

List of Figures

Figure 1 – Example of point anomalies in time series data, highlighting local (O1) and global outliers (O2) (BLáZQUEZ-GARCíA et al., 2021). 17

Figure 2 – Example of subsequence anomalies, showing local (O1) and global (O2) deviations in time series data (BLáZQUEZ-GARCíA et al., 2021). . . . 17

Figure 3 – Example of a time series anomaly in a multivariate dataset, where Variable 4 significantly deviates from others (BLáZQUEZ-GARCíA et al., 2021). 17

Figure 4 – The six types of signals. **(a)** Type 1 signal represented by a single sinusoidal wave with constant frequency, amplitude and phase. **(b)** Type 2 signal represented by a linear combination of several type 1 signals. **(c)** Type 3 signal represented by a single component with constant amplitude and phase, but with a time-varying frequency. **(d)** Type 4 signal represented by a linear combination of several type 3 signals. **(e)** Type 5 signal represented by a type 4 signal with additive and multiplicative white noise. **(f)** Type 6 signal represented by a sinc function with additive white noise. (BOASHASH, 2016c). 19

Figure 5 – Time and frequency representations of the signals defined in Equations 2.9 and 2.10. **(a)** Time and frequency representations of the signal x_1. **(b)** Time and frequency representations of the signal x_2. (BOASHASH, 2016c) 22

Figure 6 – Structure of a neural network. (***a***) Fully connected perceptrons arranged in layers (P., 2025). (***b***) Perceptron with activation function (WEAVER, 2024). 28

Figure 7 – Illustration of the convolution operation in a convolutional neural network. (RIEBESELL, 2022) 31

Figure 8 – Example of max pooling operation in a convolutional neural network. (VAART; LAMBERS, 2019) 32

Figure 9 – Architecture of a recurrent neural network showing the recurrent connections between hidden states. Adapted from (SCHMIDT, 2019) . . . 33

Figure 10 – Structure of an LSTM cell with input, forget, and output gates. Adapted from (WEI et al., 2021) 35

Figure 11 – Diagram of the attention mechanism showing query, key, value vectors, and the computation of the context vector. (NIU; ZHONG; YU, 2021) 37

Figure 12 – Architecture of the vanilla Transformer model, with encoder and decoder components. (LIN et al., 2021) 38

Figure 13 – Main railway track components, including superstructure and substructure. (KAEWUNRUEN[1]; REMENNIKOV[1], 2008) 41
Figure 14 – Example of how velocity differences affect acceleration measurements on the same track segment. (***a***) shows the two velocity profiles, A and B. (***b***) illustrates the corresponding acceleration measurements for each profile. Adapted from (ONO et al., 2023). 46
Figure 15 – Reduction of false positives in anomaly detection after velocity correction. (***a***) shows the heatmap of the acceleration measurements before correction, while (***b***) shows the heatmap after correction. Adapted from (ONO et al., 2023). 46
Figure 16 – Effect of train velocity on measured acceleration compared with simulation results and the linear and exponential model fitted to the maximum acceleration values with their corresponding R^2 values (BALOUCHI; BEVAN; FORMSTON, 2021). 47
Figure 17 – Effect of wagon mass on the relationship between acceleration and velocity for freight trains. The data points represent the 99th percentile of the absolute maximum vertical acceleration for each velocity class, colored by wagon mass class. 48
Figure 18 – 51
Figure 19 – Reconstruction of lateral irregularities using the Frequency Response Function method (De Rosa; ALFI; BRUNI, 2019). 53
Figure 20 – Data processing pipeline for the creation of the simulated dataset (TSUNASHIMA; YAGURA, 2024). 54
Figure 21 – Track irregularity prediction using Gaussian Process Regression for different vehicle speeds (TSUNASHIMA; YAGURA, 2024). 54
Figure 22 – Regression results comparing predicted and actual track irregularities from real measurement data. (TSUNASHIMA; YAGURA, 2024) 55
Figure 23 – Model architecture utilized to reconstruct the track irregularity (HAO et al., 2023). 56
Figure 24 – Reconstruction results comparing the proposed model (red) with two simpler models (blue and green) and the original measurement (black). (***a***) and (***b***) show vertical and lateral, respectively, irregularities reconstruction for the NHHSR line. (***c***) and (***d***) show the vertical and lateral, respectively, irregularities reconstruction for the BGHSR line (HAO et al., 2023). 57
Figure 25 – Dataset distribution of standard deviation of the cross level and lateral track irregularities with defined safety thresholds. Class 1 corresponds to normal conditions, while Class 2 corresponds to sections requiring maintenance intervention (ROSA et al., 2021). 59

Figure 26 – Classification results of the three classifiers on the test set. (***a***) the decision boundary for the decision tree. (***b***) the decision boundary for the linear SVM. (***c***) the decision boundary for the Gaussian SVM. (ROSA et al., 2021) 59
Figure 27 – Flowchart of the proposed methodology. 61
Figure 28 – Mass-spring-damper model of the wagon. 75

List of Tables

Table 1 – Summary of the parameters for each track class. (PIRES et al., 2021) . 42
Table 2 – Limit values for the track geometry parameters. (ABNT, 2020) 43
Table 3 – Comparison of the accuracy indices. 52
Table 4 – Evaluation metrics for different models and different lines (HAO et al., 2023). 57
Table 5 – Search space for the SIMPACK simulation. 61
Table 6 – Schedule of tasks from September to December 63

List of abbreviations and acronyms

CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
AE	Autoencoder
GAN	Generative Adversarial Network
MSE	Mean Squared Error
FRA	Federal Railroad Administration
ABNT	Associação Brasileira de Normas Técnicas
PSD	Power Spectral Density
IRV	Instrumented Railway Vehicle
TGC	Track Geometry Car
UIC	International Railway Union
MLP	Multilayer Perceptron
VAMPIRE	Vehicle dynamics simulation software
SVM	Support Vector Machine
FRF	Frequency Response Function
UIO	Unknown Input Observer
CV	Computer Vision
NLP	Natural Language Processing
ReLU	Rectified Linear Unit
GANs	Generative Adversarial Networks
EFVM	Estrada de Ferro Vitória a Minas

Contents

1 INTRODUCTION . . . 13

Introduction . . . 13

2 LITERATURE REVIEW . . . 15
2.1 Time Series Anomaly Detection . . . 15
2.1.1 Time Series . . . 15
2.1.2 Time Series Anomalies . . . 16
2.2 Basics of Signal Analysis . . . 18
2.2.1 Types of Signals . . . 18
2.2.2 Fourier Transform . . . 20
2.2.3 Power Spectral Density . . . 21
2.2.4 Limitations of Time and Frequency Representations . . . 21
2.2.5 Analytic Signals and the Hilbert Transform . . . 23
2.2.6 Instantaneous Frequency . . . 24
2.3 Time Frequency Distributions . . . 25
2.3.1 Short-Time Fourier Transform . . . 26
2.3.2 Wavelet Transform . . . 26
2.3.3 Quadratic Time-Frequency Distributions . . . 26
2.4 Time Series Anomaly Detection Models . . . 26
2.5 Basics of Deep Learning Models . . . 27
2.5.1 Neural Networks . . . 27
2.5.2 Backpropagation . . . 29
2.5.3 Convolutional Neural Networks . . . 30
2.5.4 Recurrent Neural Networks . . . 32
2.5.5 LSTM . . . 34
2.5.6 Attention . . . 35
2.5.7 Transformers . . . 37
2.6 Reconstruction-Based Models . . . 39
2.7 Railway Components . . . 40
2.8 Track Irregularities Standards . . . 41
2.8.1 FRA Standard . . . 41
2.8.2 Brazilian Standard . . . 42
2.9 How to Generate Track Irregularities from PSDs . . . 43
2.10 Assessing Railway Track Condition . . . 44
2.11 Effect of the Velocity on Train Dynamics . . . 45

2.12 Effect of Mass on Train Dynamics . . . 47
2.13 Normalization of Signals for Different Operational Conditions . . . 49
2.14 Assessing Track Irregularities from Acceleration Data . . . 51
2.15 Thresholding Acceleration Data . . . 58

3 METHODOLOGY . . . 60
3.1 Defining the Problem . . . 60
3.2 Search Space of the SIMPACK Simulation . . . 61
3.3 Machine Learning Model . . . 62

4 SCHEDULE . . . 63

BIBLIOGRAPHY . . . 64

APPENDIX 70

APPENDIX A – DERIVING THE BACKPROPAGATIONG RULE . 71

APPENDIX B – PROOF OF ANALYTIC SIGNAL THEOREM . . 72

APPENDIX C – REAL SIGNALS AND TFDS . . . 73

APPENDIX D – SPECTRAL REPRESENTATION METHOD . . . 74

APPENDIX E – AN THEORETICAL APPROACH FOR THE DEPENDANCE OF VELOCITY AND MASS . . . 75
E.1 Lightly Damped System . . . 79
E.2 Exact Evaluation at $w = 4$ . . . 80

APPENDIX F – PROOF OF THE E.0.1 LEMMA . . . 81

1 Introduction

Railways are one of the most important modes of transportation worldwide. It is known for its efficiency, cost-effectiveness, and ability to move large volumes of goods and passengers over long distances. According to the Internacional Railway Union (UIC), in 2024 railways transported more than 24 billion passengers and over 9 billion tons of freight (UIC, 2024). Given this scale, it is crucial to ensure the safety and reliability of the railway infrastructure, which can be achieved through regular maintenance and monitoring of the tracks.

The monitoring of railway tracks aims to identify potential defects that require maintenance. Track irregularities are deviations from the ideal geometry of the track that can affect the comfort and safety of train operations. They are caused by various factors, such as wear and tear, that degrade the track condition. In severe cases, they can cause accidents, such as derailments. Traditionally, monitoring of railway tracks has been done using Track Geometry Cars (TCGs) that measures directly the track geometry. In the best case scenario, this process is done monthly, due to its high operational cost and time requirements, as it requires the railway operation to halt services during inspections which can take several hours depending on the length of the track measured.

An alternative approach for monitoring railway track is to use an Instrumented Railway Vehicle (IRV), equipped with a series of sensors that collects data while the train is in operation. The IRV measures the dynamic response of the wagon due to track excitations. This method is less expensive and can be done more frequently, as it collects data from a moving vehicle. However, it comes with a trade-off as the collected data is not the geometry of the track, but the dynamic response of the wagon, which needs to be mapped to track irregularities. This mapping, however, is not straightforward, as the measured dynamic of the system is heavily influenced by various factors, such as train speed, track condition, wagon characteristics, and the inherent noise of real-world data. Therefore, IRV data must be carefully processed and analyzed to extract meaningful information about the track condition.

Several techniques have been proposed to try solving this mapping problem, such as the Kalman filter or traditional machine learning methods. These methods aim to reconstruct the track irregularities from the data collected by the IRV, but often face limitations when dealing with complex and noisy data. More recently, to deal with the complexity of the problem, deep learning methods have been proposed as a solution to reconstruct track irregularities. These models use advanced architectures to learn the correlations and patterns within the data. However, most of the proposed models don't

consider the influence of external factors, such as train speed and wagon mass, on the measured responses. This can reduce the robustness of the method, as it might not be able to generalize well to different operational conditions.

This thesis proposes a unsupervised deep learning model capable of reconstructing railway vertical track irregularities from acceleration measurements. Data from a multibody simulation was used to train the machine learning model and real-world data was used to validate it. The study considers the effects of train velocity and wagon mass on the measured responses, proposing a correction method that normalizes the acceleration under different operational conditions. By combining validated multibody simulations with advanced learning algorithms, the proposed approach aims to develop a robust framework for railway track monitoring. Unlike state-of-the-art methods, it explicitly incorporates the influence of velocity and mass into the acceleration data used for training the deep learning model.

This work is organized as follows: Section 2 presents a literature review on the topic of railway track monitoring and methods used to reconstruct track irregularities from acceleration data. Section 3 describes the methodology used in this study, including the defined parameters of the simulation and the machine learning model used. Section **??** presents the results obtained, and Section 4 shows a schedule of activities to be performed in this thesis.

2 Literature Review

2.1 Time Series Anomaly Detection

In recent years, rapid technological advancements have led to a significant increase in the volume of data collected and generated from diverse sources such as sensors, databases, and files (ATTOH-OKINE, 2017). This large volume of data is now commonly referred to as Big Data. A substantial portion of Big Data consists of data points collected in sequential time steps, also known as time series. Analyzing time series data presents a complex set of challenges, such as: (TSAI et al., 2015).

- **Large Volume of Data**: Time series data can be collected at high frequencies, leading to large datasets that consume significant storage and processing resources.
- **High Dimensionality**: Data can be collected by multiple sensors, resulting in high-dimensional datasets, that, sometimes, can have multiple redundant variables.
- **Variables Correlations**: In multivariate time series, multiple variables might have complex correlations, making it challenging to model their relationships.

One of the most studied problems in this area is anomaly detection. It consists of identifying outliers or abnormal patterns within data and has a variety of applications, from financial systems and medical diagnosis to urban management and fault detection (SAMARIYA; THAKKAR, 2021).

The following sections will dive deeper into the concept of a time series, the different natures of anomalies in data, and some of the models used to detect these anomalies.

2.1.1 Time Series

A time series (TS) can be defined as a set of points indexed sequentially over time and is often divided into univariate and multivariate. A univariate TS is a set of real values from a single variable that changes over time $\mathbf{X} = \{x_t\}$ for $t \in T$, where T is the total time. On the other hand, a multivariate TS is a set of real values from multiple variables that change over time, i.e., $\mathbf{X} = \{\mathbf{x}_t\}$ for $t \in T$, where $\mathbf{x}_t$ is a k-dimensional vector defined by $\mathbf{x}_t = (x_{1t}, ..., x_{kt})$ and each x_{it} is a univariate TS.

A subsequence of length $n \leqslant T$ can be defined as $\mathbf{S} = \{\mathbf{x}_{p:p+n}\}$ from the time series $\mathbf{X}$, where $p \in T$ and $p \leqslant T - n + 1$ (BLáZQUEZ-GARCíA et al., 2021).

Time series data exhibit dependencies that vary based on their type. In univariate time series, temporal dependency occurs when the value at time t, x_t, is influenced by its preceding values, $x_{p:t}$. In multivariate time series, in addition to temporal dependencies, there are spatial dependencies, representing the correlations between different variables within the same time step (DARBAN et al., 2024).

2.1.2 Time Series Anomalies

According to Grubbs (GRUBBS, 1969) an anomaly can be defined as an outlier that deviates greatly from the general distribution of data. These outliers can be a single observation or a subsequence of the whole time series and they can be further distinguished as a local outlier, which deviates from the local behavior of the data, or a global outlier, which deviates from the global behavior of the data. Building on this, Blázquez-García et al. (BLáZQUEZ-GARCíA et al., 2021) divides anomalies into three main categories:

1. *Point Anomalies*: A point outlier can be defined as a single observation that deviates from the global behavior (global anomaly) or from the behavior of it's neighborhood (local anomaly). This type of anomaly can be identified using equation 2.1

$$|x_t - \hat{x}_t| > \tau \tag{2.1}$$

   where $\hat{x}_t$ is the expected result, x is the original data point and τ is the threshold. If the difference is greater than the threshold τ, then x_t is identified as an anomaly. Figure 1 shows an example of this anomaly, where O1 is a local outlier and O2 is a global anomaly.

2. *Subsequence Anomaly*: A subsequence outlier refers to a consecutive portion of points from the time series whose joint behavior is abnormal. It is important to note that each of these data points alone doesn't need to be a point anomaly, but together they form an anomaly. Similarly to the point outliers, the subsequence outliers can be global or local. These anomalies can be identified by the equation 2.2:

$$diss\left(C, \hat{C}\right) > \tau \tag{2.2}$$

   where C is the the actual cycle or shape of the subsequence, $\hat{C}$ is the expected output, τ is the threshold and $diss$ is a function that computes the dissimilarity between the two subsequences, such as the cosine similarity. Figure 2 shows an example of this, where O1 is a local anomaly and O2 is a global anomaly.

3. *Time Series Anomaly*: A time series anomaly is an entire time series that presents unusual behavior. Note that this kind of anomaly can only be detected in multivariate time series, as it needs to consider the behavior of one feature compared to the

others. Figure 3 shows an example of this problem, where Variable 4 behavior differs greatly from the other 3.

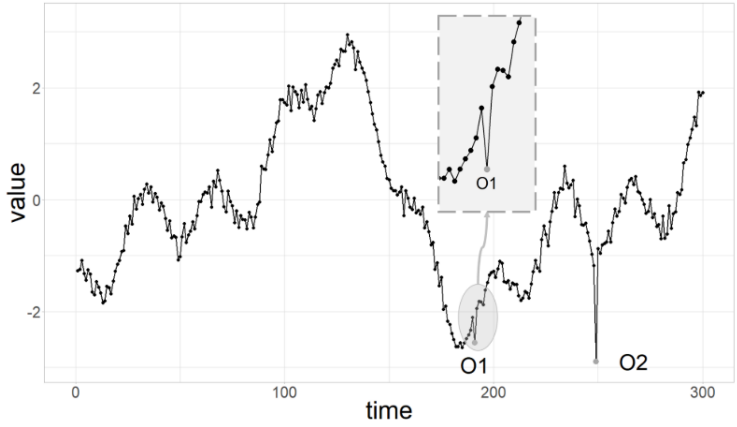


Figure 1 – Example of point anomalies in time series data, highlighting local (O1) and global outliers (O2) (BLáZQUEZ-GARCíA et al., 2021).

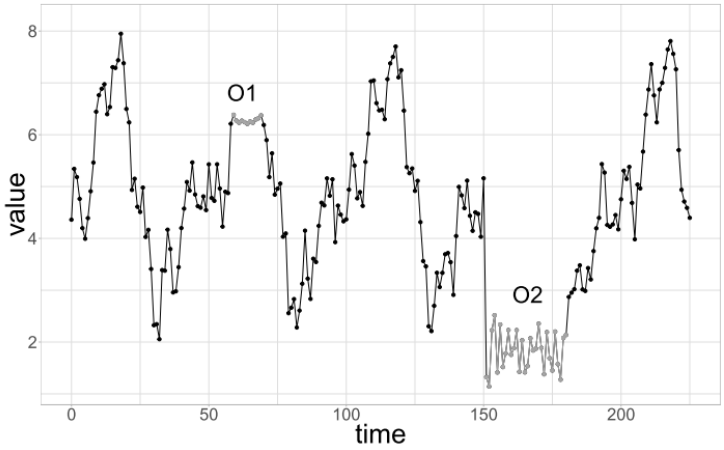


Figure 2 – Example of subsequence anomalies, showing local (O1) and global (O2) deviations in time series data (BLáZQUEZ-GARCíA et al., 2021).

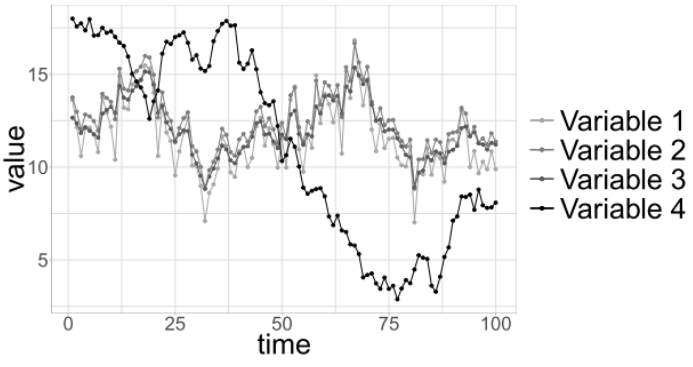


Figure 3 – Example of a time series anomaly in a multivariate dataset, where Variable 4 significantly deviates from others (BLáZQUEZ-GARCíA et al., 2021).

2.2 Basics of Signal Analysis

In this section the basics of signal analysis will be presented. The main idea behind this section is to provide the theoretical background that will be used in the next chapters, specially when dealing with mass and velocity effects on acceleration data.

(No futuro escrever melhor essa parte aqui, talvez seja interessante adicionar ou remover algumas sessoes se essas nao forem utilizadas pela tese.)

2.2.1 Types of Signals

In Section 2.1.1 a time sereis was defined as a sequence of data points indexed sequentially over time. A time series can be considered a signal, which is a function that conveys information about the behavior of a system, such as an acoustic wave from a speech, an eletronic wave carrying information of an image or, in our case, the output of an acceleromter sensor (PRIEMER, 1991). Signals can be classified into six different types depending on the level of complexity (BOASHASH, 2016c):

1. **Type 1 Signal**: This type of signal represents a simple and deterministic sinusoidal wave, described by a constant frequency f_c, a constant amplitude and phase. This type of signal is shown in Figure 4(a).

2. **Type 2 Signal**: This type of signal is a multicomponent signal, which is a liner combination of several type 1 sinusoidal waves, each with a different frequency, amplitude and phase. This type of signal is illustrated in Figure 4(b).

3. **Type 3 Signal**: This type of signal is comprised of a single component with constant amplitude and phase, but with a time-varying frequency $f_i(t)$. This type of signal is shown in Figure 4(c). One example of this is the frequency modulated (FM) signal used in telecommunication systems. The dominant frequency of this signal at time t is called the instantaneous frequency (IF) and will be discussed in more detail in Section 2.2.6.

4. **Type 4 Signal**: This type of signal is a multicompening signal composed of multiple type 3 signals summed together, each with a different time-varying frequency, amplitude and phase. This type of signal is illustrated in Figure 4(d).

5. **Type 5 Signal**: A type 5 signal is a type 4 signal with an white noise added to the sum of the components and a white noise added to the amplitude of each component. This type of signal is shown in Figure 4(e). The first type of noise is called an additive noise, while the second is called a multiplicative noise.

6. **Type 6 Signal**: The type 6 signal is a sinc function with additive withe noise added

to it. Figure 4(f) shows an example of this type of signal. This type of signal is the most important for this dissertation, as it reveals possible defects on the railway line.

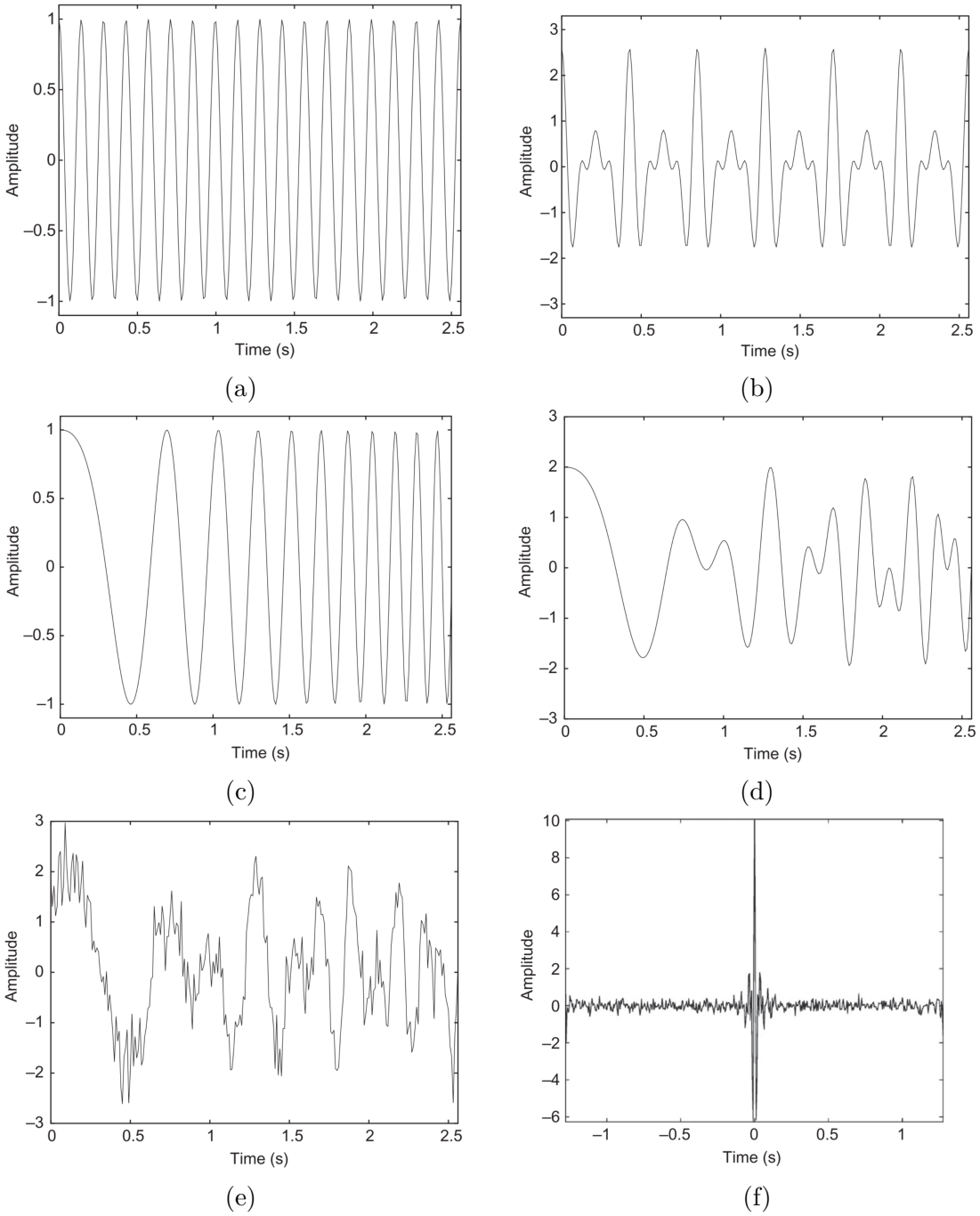


Figure 4 – The six types of signals. **(a)** Type 1 signal represented by a single sinusoidal wave with constant frequency, amplitude and phase. **(b)** Type 2 signal represented by a linear combination of several type 1 signals. **(c)** Type 3 signal represented by a single component with constant amplitude and phase, but with a time-varying frequency. **(d)** Type 4 signal represented by a linear combination of several type 3 signals. **(e)** Type 5 signal represented by a type 4 signal with additive and multiplicative white noise. **(f)** Type 6 signal represented by a sinc function with additive white noise. (BOASHASH, 2016c).

It is important to highlight that an algorithm that extracts characteristics from a signal changes with the type of signal, meaning that an algorithm that works well for a type 1 signal might not work well for a type 4 signal (BOASHASH, 2016c).

2.2.2 Fourier Transform

In Section 2.2.1, the six different types of signals and their representation in the time domain were presented. However, signals can also be represented in the frequency domain. The Fourier Transform (FT) is a mathematical tool that maps a time-domain signal to its frequency-domain representation by decomposing it into its constituent frequency components. Formally, the FT is a linear operator defined on suitable function spaces. For example, if $f(t) \in L^1(\mathbb{R})$, where $L^1(\mathbb{R})$ denotes the Banach space of absolutely integrable functions, the Fourier Transform is defined as (GROCHENIG, 2001)

$$\mathcal{F}[f](\omega) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i \omega t}\, dt, \tag{2.3}$$

where ω denotes the frequency variable. Since the Fourier Transform is invertible on appropriate function spaces, the original time-domain signal can be recovered from its frequency-domain representation using the inverse Fourier Transform (IFT), defined as

$$f(t) = \mathcal{F}^{-1}[F](t) = \int_{-\infty}^{\infty} F(\omega) e^{2\pi i \omega t}\, d\omega, \tag{2.4}$$

where $F(\omega) = \mathcal{F}[f](\omega)$ is the Fourier Transform of $f(t)$. Notice that to get back the signal to the time domain, the IFT integrates over all frequencies, including negative ones. The mathematical explanation for this is that if $f(t)$ is a real-valued signal, then its FT satisfies the property $F(\omega) = F^*(-\omega)$, where $F^*(\omega)$ denotes the complex conjugate of $F(\omega)$, i.e., its FT has a Hermitian symmetry. This means that the negative frequencies are duplicates from the positive ones (BOASHASH, 2016c).

Another important result to discuss is the Parseval's theorem. It states that if $f \in L^1 \cap L^2(\mathbb{R})$, then:

$$E_s = \int_{-\infty}^{\infty} |f(t)|^2\, dt = \int_{-\infty}^{\infty} |\mathcal{F}[f](\omega)|^2\, d\omega \tag{2.5}$$

where E_s is the energy of the signal (GROCHENIG, 2001). This means that the energy of the signal is preserved by the FT, which allows for the identification of frequencies that contribute the most to the signal's energy.

2.2.3 Power Spectral Density

In Section 2.2.2, it is shown that taking the FT of a signal preserves the total energy of the signal, which allows for the identification of the frequencies that contribute the most to the signal's energy. However, this only applies to absolutely integrable functions, that is, functions whose total energy E is finite. This does not hold, however, for a general stationary stochastic signal, whose energy is usually infinite (KAY; MARPLE, 1981). In this case, it is more useful to analyze the distribution of power (energy per unit of time) across the frequency spectrum, which is captured by the Power Spectral Density (PSD).

Formally, for a signal $x(t)$, the PSD is defined as (YOUNGWORTH; GALLAGHER; STAMPER, 2005)

$$S_x(\omega) = \lim_{T\to\infty} \frac{1}{T} \left|X_T(\omega)\right|^2, \tag{2.6}$$

where $X_T(\omega)$ denotes the Fourier Transform of the signal truncated to an observation window of length T. This truncation is necessary so that the definition remains valid for signals that do not have a formal Fourier Transform.

An important result is the Wiener–Khinchin theorem, which relates the PSD to the signal's autocorrelation function $R_x(\tau)$ (KAY; MARPLE, 1981),

$$S_x(\omega) = \int_{-\infty}^{\infty} R_x(\tau)\, e^{-i2\pi\omega\tau}\, d\tau, \tag{2.7}$$

where $R_x(\tau)$ is defined as:

$$R_x(\tau) = \mathbb{E}[x(t+\tau)x^*(t)]. \tag{2.8}$$

2.2.4 Limitations of Time and Frequency Representations

As discussed in Section 2.2.2, the FT can convert a signal from the time domain to the frequency domain. However, relying only on these two representations can be limiting, as the variables t and ω are treated independently, i.e., to obtain one, the other is integrated out, which causes the representations to be nonlocalized w.r.t. the excluded variable (BOASHASH, 2016a). For example, taking the FT of a signal $f(t)$, the resulting frequency-domain representation $F(\omega)$ provides information about the frequency content of the signal but does not provide any information about when those frequencies occur in time. This limitation is particularly problematic for non-stationary signals, where the frequency content changes over time. This is clearly shown by Boashash (BOASHASH, 2016c) using the following two signals:

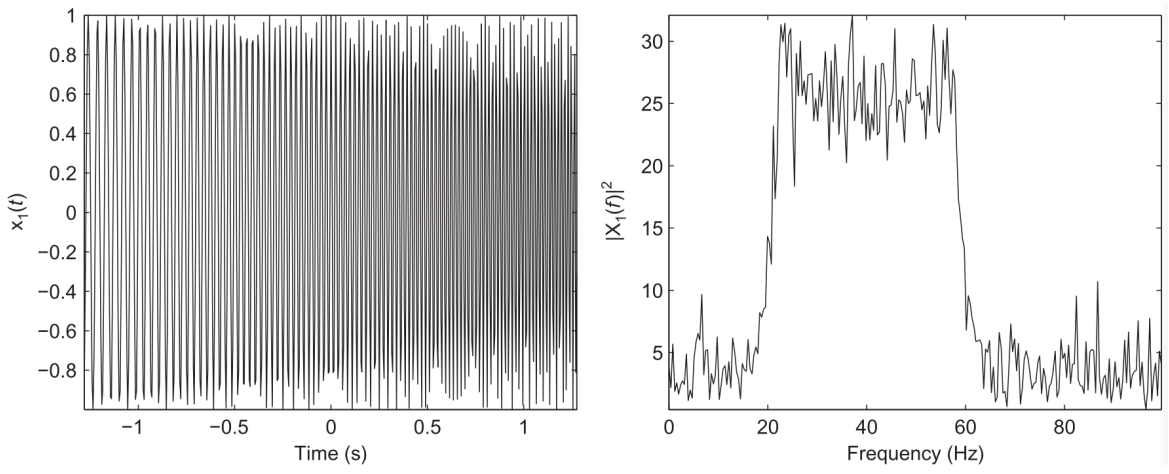
$$x_1(t) = \text{rect}\left(\frac{t - T/2}{T}\right) \cos\left[2\pi\left(f_0 t + \alpha\frac{t^2}{2}\right)\right] + N_1(t) \tag{2.9}$$

$$x_2(t) = \text{rect}\left(\frac{t - T/2}{T}\right) \frac{\text{sen}\,(\pi B t)}{\pi B t} \cos\left(2\pi f_c t\right) + N_2(t) \tag{2.10}$$

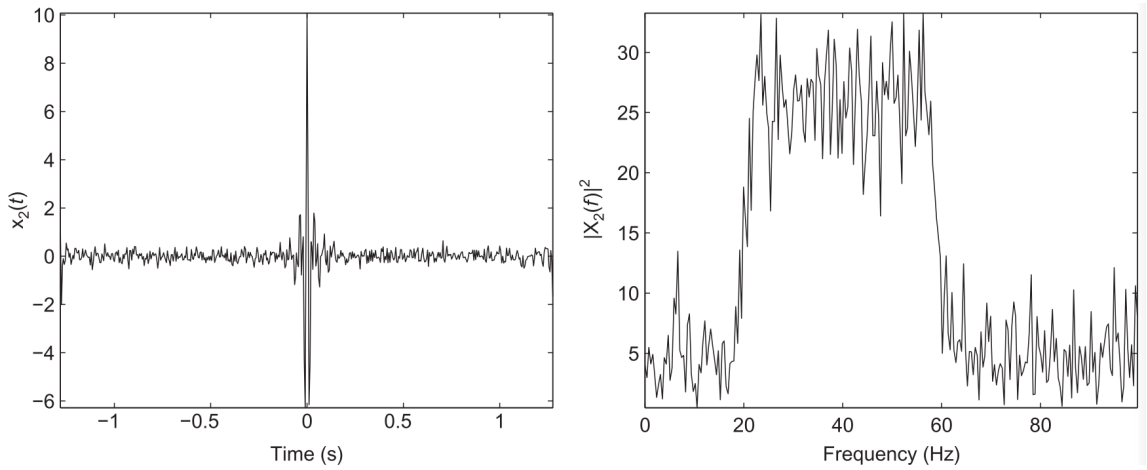
where rect(·) is the function

$$\text{rect}(x) = \begin{cases} 1, & |x| \leqslant \frac{1}{2} \\ 0, & |x| > \frac{1}{2} \end{cases} \tag{2.11}$$

f_0 is the initial frequency, $\alpha = B/T$ is the frequency sweep, $f_c = f_0 + B/2$ is the center of the bandwidth, $N_1(t)$ and $N_2(t)$ are additive noise terms, and T is the duration of the signals, such that $BT \gg 1$. The time and frequency representations of the x_1 and x_2 signal are shown in Figures 5(a) and 5(b), respectively. As it can be seen, both have similar frequency representations, but their time representations are very different. Futhermore, taking a look at the time representations, it is hard to identify which frequencies dominate at which time, or even if there is more than one frequency present. It is also difficult to identify the evolution of the frequencies over time, a characteristic of nonstationary signals. This is a limitation of the classical representations of the signals. In the Section 2.3, a more complex time-frequency representation will be presented.



(a)



(b)

Figure 5 – Time and frequency representations of the signals defined in Equations 2.9 and 2.10. **(a)** Time and frequency representations of the signal x_1. **(b)** Time and frequency representations of the signal x_2. (BOASHASH, 2016c)

2.2.5 Analytic Signals and the Hilbert Transform

In Section 2.2.2 it is discussed about negative frequencies and the Hermitian symmetry of real-valued signals. More than that, a signal $f(t)$ is said to be real if and only if:

$$F(\omega) = F^*(-\omega) \tag{2.12}$$

where $F(\omega)$ is the FT of $f(t)$. In this case, the negative part of the frequencies can be discarded withouth the loss of information. In fact, it allows the signal to be sampled at half the Nyquist rate (CLAASEN; MECKLENBRäUKER, 1980). This duplication of information allows the negative frequencies to be discarded without losing of information, which forms an analytic signal. An analytic signal $z(t)$ is a complex-valued signal that contains only positive frequency components, that is:

$$Z(\omega) = 0, \quad \text{for } \omega < 0 \tag{2.13}$$

Using this definition, it is possible to state the following theorem (BOASHASH, 2016a):

Theorem 2.2.1. *Consider two real signals $s(t)$ and $y(t)$. The signal*

$$z(t) = s(t) + iy(t) \tag{2.14}$$

is analytic with real DC component if and only if

$$Y(\omega) = -isgn(\omega)S(\omega) \tag{2.15}$$

where $S(\omega)$ and $Y(\omega)$ are the Fourier transforms of $s(t)$ and $y(t)$, respectively, and

$$sgn(\omega) = \begin{cases} 1, & \omega > 0 \\ 0, & \omega = 0 \\ -1, & \omega < 0 \end{cases} \tag{2.16}$$

The proof of this theorem is presented in Appendix B.

If $y(t)$ nd $s(t)$ are related by Equation 2.15, then $y(t)$ is called the Hilbert transform (HT) of $s(t)$ and denoted as:

$$y(t) = \mathcal{H}\{s(t)\}. \tag{2.17}$$

Using the HT, it is possible to construct the analytic signal $z(t)$:

$$z(t) = s(t) + i\mathcal{H}\{s(t)\}. \tag{2.18}$$

In this case,$z(t)$ is called the analytic associate of $s(t)$.

By taking the IFT of Equation 2.15, combining it with Equation 2.17 and using the convolution property of the FT, it is possible to show that the HT can be defined as:

$$\mathcal{H}\{s(t)\} = \frac{1}{\pi t} * s(t) = \frac{1}{\pi}\text{p.v.}\int_{-\infty}^{\infty}\frac{s(\tau)}{t-\tau}d\tau \tag{2.19}$$

The reasons for using $z(t)$ and the HT instead of $s(t)$ will be highlighted in Section 2.3.

2.2.6 Instantaneous Frequency

As stated in Section 2.2.5, given a real-valued signal $s(t)$, it is possible to construct an analytical signal $z(t) = s(t)+i\mathcal{H}\{s(t)\}$, where $\mathcal{H}\{s(t)\}$ is the Hilbert transform of $s(t)$. This analytic signal can be expressed as:

$$z(t) = a(t)e^{i\phi(t)}, \tag{2.20}$$

where $a(t)$ is the amplitude envelope and $\phi(t)$ is the phase of the signal. The instantaneous frequency (IF) of $s(t)$ is then defined as the rate of change of this phase, $f_i(t) = \frac{1}{2\pi}\frac{d\phi}{dt}$, extending the notion of frequency to signals whose spectral content may vary over time.

As an example, consider the case of a pure cosine, $x(t) = A\cos(2\pi f_0 t)$, whose analytic signal has phase

$$\phi(t) = 2\pi f_0 t. \tag{2.21}$$

The phase here increases linearly with time, at a constant rate of $2\pi f_0$ radians per second. Since a phase advance of $2\pi f_0$ radians per second is equivalent to f_0 complete cycles per second, the rate of change of phase coincides with the frequency of the signal in this elementary case. This observation does not depend on the phase being linear in time (STANKOVIć et al., 2014). When $\phi(t)$ advances more quickly during some intervals and more slowly during others, the signal oscillates faster or slower accordingly, even though a single frequency no longer characterizes the waveform as a whole.

The definition above, however, is subject to controversy. For example, Shekel (SHEKEL, 1953) argues that IF defined in Equation 2.21 is not a unique function of time because, for example, a amplitude modulated (AM) signal can be defined as $a(t)e^{i2\pi f_0 t}$, with $a(t)$ being a complicated function, or as $a_0 e^{i\phi(t)}$, with a_0 being a constant and $\phi(t)$ being a complicated phase function. Altough the method of creating the analytic signal described in Section 2.2.5 is a unique way to define the phase of a signal, the physical meaning of the HT, $a(t)$ and $\phi(t)$ is not always meaningful and representative. Boashash (BOASHASH, 1992) gives the conditions for this transformation to be representative of the physical phenomena, but this discussion is beyond the scope of this dissertation.

This restriction becomes more apparent when the definition is applied to a multicomponent signal, formed by the superposition of two or more oscillatory components

with distinct frequencies present simultaneously. In this case, each component of the signal will have its own instantaneous frequency and, in general, the IF of the sum of two signals is not the sum of the IFs of the individual components (BOASHASH, 2016a). In fact, $f_i(t)$, as defined above, behaves as a kind of instantaneous average, a weighted combination of the frequencies of the individual components, weighted by their relative contributions to the signal's energy at that instant (COHEN; LEE, 1988; BOASHASH, 1992). The resulting value may fall between the true component frequencies, or even outside their range altogether, and therefore cannot be interpreted as the frequency of any physical oscillation.

This limitation motivates a reinterpretation of instantaneous frequency in terms of energy distribution over time and frequency. Rather than describing the frequency of the signal at time t, $f_i(t)$ is more appropriately understood as an estimate of the frequency around which most of the signal's energy is concentrated at that instant, when the signal is examined through a joint time-frequency distribution (STANKOVIĆ et al., 2014). This concept will be discussed further in Section 2.3, where time-frequency distributions will be presented.

2.3 Time Frequency Distributions

In this section, some of the most important time-frequency distributions (TFDs) are presented. The core idea behind TFDs is to provide a joint representation of a signal such that the area under the TFD surface corresponds to the energy of the signal, that is, the TFD is an energy density function (BOASHASH, 2016b). For a monocomponent signal, the ideal TFD would concentrate all the energy of the signal along its IF (STANKOVIC; STANKOVIC, 1995). Mathematically, this can be expressed as:

$$ITF(t,\Omega) = 2\pi A^q(t)\delta(\Omega - \phi'(t)) = A^q(t)\int_{-\infty}^{\infty} e^{i\phi'(t)\tau}e^{-i\Omega\tau}d\tau. \tag{2.22}$$

However, as will be shown in the next sections, in practice, the TFDs only approximate this ideal behavior, and the energy of the signal is spread around the IF.

In Sections 2.3.1 and 2.3.2, the so-called linear TFDs will be presented. These TFDs try to represent the signal as a linear combination of elementary signals, called atoms, which are localized in time and frequency. In Section 2.3.3, the quadratic TFDs will be presented. Quadratic TFDs distribute the energy of the signal in the time-frequency plane. As will be shown, these TFDs are all based on the Wigner-Ville distribution (WVD).

Finally, another important point to highlight is that in the next sections the TFDs will be presented using the analytic conjugate of the signal, as using the real signal can lead to some problems. Check Appendix C for more details.

2.3.1 Short-Time Fourier Transform

Given a signal $s(t)$ and a real, even window function $w(t)$, the Short-Time Fourier Transform (STFT) is defined as (GROCHENIG, 2001):

$$F_s^w(t,\omega) = \int_{-\infty}^{\infty} s(\tau)w(\tau - t)e^{-i2\pi\tau\omega}d\tau, \tag{2.23}$$

The STFT

2.3.2 Wavelet Transform

2.3.3 Quadratic Time-Frequency Distributions

2.4 Time Series Anomaly Detection Models

Since the 1960s, many models for anomaly detection have been proposed, ranging from statistical models to recent deep learning models. The common main idea behind all of those is to identify low probability, i.e. low density, regions, classifying points in those areas as anomalies (SAMARIYA; THAKKAR, 2021).

Statistical models use statistical tests, like the χ^2 test, to identify abnormal points. Clustering-based models use cluster analysis to label anomalies, using, for example, the distance from the centroid of normal points or the number of points in a cluster. Distance-based models use the distance between the current time window and its neighborhood to classify anomalies. Density-based approaches use the density of the data points to find anomalies (SAMARIYA; THAKKAR, 2021). These methods described above are called traditional methods.

In recent years, with the increase of computing power, deep learning models have become increasingly prominent in anomaly detection. Unlike the traditional methods, the main advantage of deep learning models is that they are capable of identifying complex temporal and spatial correlations between variables, especially in larger multidimensional datasets, without the need for a labeled dataset (CHOI et al., 2021). Darban et at. (DARBAN et al., 2024) divides deep learning models into 3 main categories: forecasting-based models, reconstruction-based models, and representation-based models.

Forecasting-based models try to predict a future subsequence of the time series using historical data. These models learn the patterns and trends within data to anticipate what is expected to occur. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTMs) are examples of this category. Anomalous values are detected if they deviate greatly from what the model predicts.

Reconstruction-based models attempt to recreate the input timeseries based on the values of past points or windows. These models use architectures like autoencoders

(AE) and Generative Adversarial Networks (GANs). Anomalies are identified using the reconstruction error between the predicted series and the original series. The advantage of reconstruction-based models over forecasting ones is that they adapt better to rapid changes in a timeseries, which may cause it to become unpredictable. This happens because reconstruction models have access to the current timeseries data, which is not available in forecasting models (DARBAN et al., 2024). This better precision comes with a delay in the detection time of an anomaly, but, in general, these models are preferred over forecasting models.

Representation-based models focus on learning meaningful representations of the data that can be used in downstream tasks, like anomaly detection. These models use advanced architectures like transformers and self-supervised learning to extract high-level features from the data. From this representation, it is easier to solve a specific task and use simpler models (e.g., clustering or density-based methods) to identify anomalies. The downside of these models is that they are computationally expensive to train, often needing a large amount of data.

An important point to highlight is that the majority of the models described above are unsupervised models. An unsupervised model is a model that can learn patterns from data without the need for a labeled dataset. This flexibility is desired because of the inherently unlabeled nature of historical data and the unpredictability of anomalies, which makes it harder to define an anomaly, especially if anomalies are rare. In this work, the focus will be on unsupervised reconstruction models.

Another important point to note is that, as will be detailed in Sections 2.14 and 3, the model proposed in this thesis is not designed specifically for anomaly detection. Instead, its primary objective is to learn how to reconstruct track irregularities from acceleration data. Despite that, the main idea that motivates the model's development is closely related to anomaly detection. For this reason, a review of anomaly detection methods was included.

2.5 Basics of Deep Learning Models

Before diving deeper into anomaly detection models, the fundamental concepts of deep learning will be presented first. This preliminary discussion will provide the necessary context and theoretical basis for the subsequent exploration of reconstruction models.

2.5.1 Neural Networks

Neural networks are the most basic structure in deep learning. It is inspired by the way the human brain works (AGATONOVIC-KUSTRIN; BERESFORD, 2000).

These networks are composed of thousands, or even millions, of interconnected units called perceptrons, which mimic the behavior of biological neurons. In a neural network, these perceptrons are arranged into multiple layers, as shown in Figure 6(a). In this illustration, each circle represents a single perceptron.

The neural network is divided into three main components:

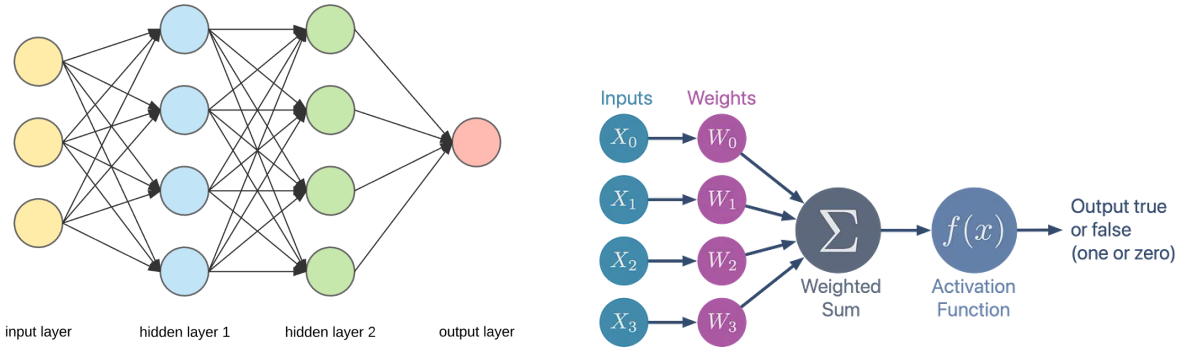
- The input layer: this layers receives raw data that will be precessed;
- The hidden layers: positioned between the input and output layers, these intermediate layers perform feature extraction;
- The output layer: this layer provides predictions or classifications

Each perceptron in a neural network processes input data linearly using the dot product. Given, then, an input array $\mathbf{x}$, the perceptron computes the output by multiplying $\mathbf{x}$ with an array of weights $\mathbf{w}$ and adding a bias b, producing a scalar output o (GURNEY, 1997):

$$o = \sum_i w_i x_i + b \tag{2.24}$$

The weights and biases are learned and iteratively updated during the training process such that they minimize the errors.

However, in its basic form, a perceptron can only solve linearly separable problems. To address this limitation, modern perceptrons include an additional element: the activation function. An activation function introduces non-linearity into the model, enabling it to solve complex, non-linear problems (ROTH, 2016). For reasons that will be explained later, it is common to choose differentiable activation functions. Common activation functions include the Rectified Linear Unit (ReLU) and the hyperbolic tangent (tanh). The general structure of a perceptron, including the activation function, is illustrated in Figure 6(b).



(a) (b)

Figure 6 – Structure of a neural network. (***a***) Fully connected perceptrons arranged in layers (P., 2025). (***b***) Perceptron with activation function (WEAVER, 2024).

As mentioned earlier, the weights and biases of the perceptrons are adjusted and optimized during the training phase of a neural network. This optimization process involves a method called backpropagation, which is essential for minimizing the error in predictions. Backpropagation, discussed in detail in Section 2.5.2, leverages the gradient descent algorithm to iteratively update the network's parameters, ensuring improved accuracy over successive training iterations.

2.5.2 Backpropagation

During the training phase of a neural network, its weights and biases are systematically optimized to minimize the error. This iterative process comprises two key subprocesses: a forward pass followed by a backward pass. In a forward pass, the inputs are passed through the network and the outputs are calculated. Then, given an expected output, an error is computed using a function called the loss function. The backward pass, then, uses this error to update the weights and biases of the neural network, making it better at making right predictions. This process continues until the error is small enough.

The process of propagating the error back through the network during the backward pass is referred to as backpropagation. It can be broken down into four primary steps (CILIMKOVIC, 2015):

1. Feed foward the inputs;
2. Backpropagation to the output layer;
3. Backpropagation to the hidden layers;
4. Weights updates.

The first step is just feeding forward the inputs $\mathbf{x}$ through the network and obtaining the outputs $\mathbf{o}$. Then, a loss is computed using the loss function $\mathbf{L}$. There are a variety of loss functions, an example being the Mean Squared Error (MSE):

$$\mathbf{L} = \frac{1}{N}\sum_{i}(y_i - o_i)^2 \tag{2.25}$$

where y_i is the desired output and o_i is the network output. To backpropagate the error, the algorithm computes gradients of the loss function with respect to the parameters of the neural network. This is done using the chain rule twice (ROTH, 2016):

$$\frac{\partial \mathbf{L}}{\partial w_{ij}} = \frac{\partial \mathbf{L}}{\partial o_j}\frac{\partial o_j}{\partial net_j}\frac{\partial net_j}{\partial w_{ij}} \tag{2.26}$$

where w_{ij} is the i-th weight of the j-th perceptron, net_j is the result of the dot product before applying the activation function, i.e.,

$$net_j = \sum_{i} w_{ij}x_j \tag{2.27}$$

with x_j being the input, and o_j the perceptron output, i.e.,

$$o_j = \varphi(net_j) \tag{2.28}$$

where φ is the activation function of the perceptron. For this reason, it is necessary that the activation function is a differentiable function.

To update the weights, the algorithm computes, then, the incremental Δw_{ij} using:

$$\Delta w_{ij} = -\eta \delta_j x_{ij} \tag{2.29}$$

where η is a small positive number known as the learning rate, $\delta_j = \dfrac{\partial \mathbf{L}}{\partial net_j}$ and x_{ij} are the inputs of the j-th perceptron.

The weights are updated as:

$$w_{ij} \leftarrow w_{ij} + \Delta w_{ij} \tag{2.30}$$

For a full demonstration of these formulas, check Roth (ROTH, 2016).

Despite its effectiveness, the classical backpropagation algorithm has some problems associated with its convergence. For example, if the learning rate is too small, the convergence is very slow and can converge to a local minimum; on the other hand, if it is too large, the algorithm can diverge (RUDER, 2016). Several modifications to the original algorithm have been proposed. In Rumelhart et al. (RUMELHART; HINTON; WILLIAMS, 1986) authors propose to include a momentum term to speed up the convergence. In Tieleman (TIELEMAN, 2012) the author proposes an adaptative learning rate algorithm that divides the learning rate by the moving average of the gradients. Additionally, Kingma and Ba (KINGMA; BA, 2017) introduced the Adam optimizer, adapting the learning rate for each parameter individually.

2.5.3 Convolutional Neural Networks

Convolutional neural networks (CNN) achieved great popularity in modern machine learning, after Krizhevsky, Sutskever and Hinton (KRIZHEVSKY; SUTSKEVER; HINTON, 2012) paper. In this work, the authors proposed a deep CNN that won the ImageNet LSVRC-2010 contest, outperforming second place by a large margin.

A CNN, similarly to the neural network, was inspired by the biological neurons, but, different from the conventional neural network, where all the neurons are connected and interact with all neurons in adjacent layers, the CNN employs a sparse architecture, i.e., the neurons are connected only to a local region of the input. This sparsity reduces the amount of training parameters, which reduces the computational cost and speeds up the training process (ALZUBAIDI et al., 2021; LI et al., 2022).

CNNs are widely used in computer vision to process and analyze 2D images and are used to solve a variety of image-related problems, like classification and object detection. While this section focuses on explaining CNN concepts with the assumption of a 2D input, it is important to note that the principles discussed are equally applicable to 1D inputs, like in a timeseries.

The basic block of a CNN is the convolution operation. A convolution can be defined as the dot product between an input x of shape (m, m, r), where m is the height, which is the same as the width, and r is the depth, also called the channel number, and k convolutional filters, called kernels. A kernel is a grid of weights, with shape (n, n, q), where $n < m$ and $q \leqslant r$, that convolves with the input, producing, each, an output of size $(m - n - 1)$, called the feature maps (ALZUBAIDI et al., 2021). These kernels compute the pass value similarly to the

$$h^k = f\left(W^k * x + b_k\right) \tag{2.31}$$

where W^k is the vector of weights from the kernel, f is an activation function, and b^k is the bias. Figure 7 shows an example of this operation between a 2D binary matrix and a 3×3 kernel.

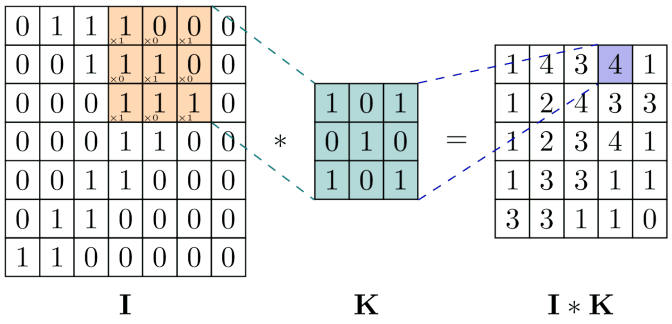


Figure 7 – Illustration of the convolution operation in a convolutional neural network. (RIEBESELL, 2022)

In addition to the convolution operation, there are two other important parameters in convolutional neural networks: the padding and the stride. Their main function is to control the output size of the convolution, helping to maintain the spatial characteristics of the input.

The padding is the process of adding additional rows and columns around the border of the input matrix, typically filled with zeros, a process known as zero padding. Without the padding, a convolution with a kernel of size $n \times n$ reduces the output size by $n-1$ in each spatial dimension. This would cause a rapid decrease in the size of the features, that would force the use of small kernels (GOODFELLOW; BENGIO; COURVILLE, 2018).

Stride, on the other hand, controls how the kernel moves across the input matrix, by skipping some of the elements from it. In a stride 1 convolution, the kernel shifts one unit at a time, covering all the positions. In a stride 2 convolution, the kernel would shift 2 elements each time, which causes the kernel to convolve with every other element. In general, a stride of size s means that the kernel will only convolve every s elements. This is done primarily to reduce computational cost, at the expense of extracting the features less finely (GOODFELLOW; BENGIO; COURVILLE, 2018).

Now with these two additional operations, the output size can be computed using the formula:

$$O = \left\lfloor \frac{I + 2 \cdot p - k}{s} \right\rfloor + 1 \tag{2.32}$$

where O is the output size, I is the input size, p is the padding, k is the kernel size, and s is the stride.

Another important feature that is widely used in CNNs is the pooling layer. Pooling layers are used to further reduce the spatial dimension of the feature map by replacing the output value of the convolution operation with a local summary statistic, which helps the model to develop representations that are invariant to small translations (GOODFELLOW; BENGIO; COURVILLE, 2018).

One of the most widely used types is max pooling, which selects the maximum value within a local neighborhood, as shown in Figure 8. Other common pooling methods include average pooling, which computes the mean of the values within the region, and weighted average pooling, where each value contributes proportionally based on predefined weights.

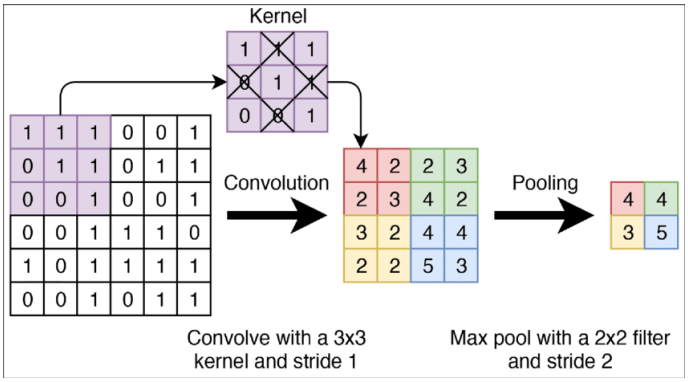


Figure 8 – Example of max pooling operation in a convolutional neural network. (VAART; LAMBERS, 2019)

2.5.4 Recurrent Neural Networks

Recurrent Neural Networks (RNNs) are a type of artificial neural network specifically designed to work with sequential data, such as time series, speech recognition, text generation, and language modeling (FANG; CHEN; XUE, 2021; ALZUBAIDI et al.,

2021). The main difference between an RNN and the networks discussed in Sections 2.5.1 and 2.5.3 is how data is processed inside the RNN. In the MLP and the CNN networks, it is assumed that each input is independent from each other, which can be a valid assumption when dealing with images, but might be a severe limitation when dealing with sequential data, like weather data, where, for example, the current temperature depends on the last few measurements.

RNNs deal with this kind of problem by introducing a cycle in their architecture allows information to persist across time steps (SCHMIDT, 2019), as shown in Figure 9. The key idea is to add a hidden-to-hidden weight matrix $\boldsymbol{W}_{hh}$ that carries information from the previous hidden state $\boldsymbol{H}_{t-1}$ in the computation of the current hidden state $\boldsymbol{H}_t$.

The current hidden state can be computed using Equation 2.33:

$$\boldsymbol{H}_t = \phi(\boldsymbol{X}_t \boldsymbol{W}_{xh} + \boldsymbol{H}_{t-1} \boldsymbol{W}_{hh} + \boldsymbol{b}_h) \tag{2.33}$$

where $\boldsymbol{X}_t$ is the input vector, $\boldsymbol{W}_{xh}$ is the input-to-hidden weight matrix, $\boldsymbol{b}_h$ is the bias, and ϕ is an activation function.

An important observation is that it is not necessary to explicitly include all previous hidden states, from $\boldsymbol{H}_0$ up to $\boldsymbol{H}_{t-2}$, when computing $\boldsymbol{H}_t$. This is because, at each iteration, the RNN updates $\boldsymbol{H}_t$ by combining the input $\boldsymbol{X}_t$ with the previous hidden state $\boldsymbol{H}_{t-1}$, which is the result of the computation between $\boldsymbol{X}_{t-1}$ and $\boldsymbol{H}_{t-2}$, and so on recursively. This way, the network can keep information about the previous time steps when computing the next one.

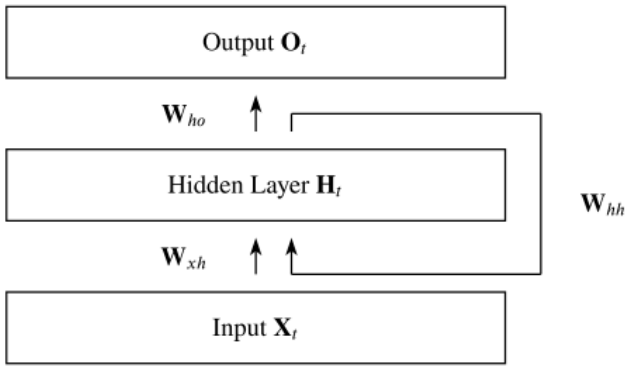


Figure 9 – Architecture of a recurrent neural network showing the recurrent connections between hidden states. Adapted from (SCHMIDT, 2019)

The main problem in the original RNN architecture arises during the training phase, specifically when computing gradients for weight updates. As gradients are propagated backward through time, a series of repeated multiplications occurs. If these values are greater than 1, the gradients can grow exponentially, causing the network weights to change abruptly fast. Conversely, if the values are less than 1, the gradients decay exponentially,

which causes the gradient to vanish (KOLEN; KREMER, 2001; GLOROT; BENGIO, 2010). The vanishing gradient then causes the network to forget information from the initial time steps, which makes it difficult for RNNs to learn long-term dependencies in data.

This inherent problem motivated the introduction of more complex architectures, such as the LSTM networks. These networks will be explained in more detail in 2.5.5, but they mitigate the vanishing gradient problem, which caused the original RNNs to be barely used in today's research (FANG; CHEN; XUE, 2021; GAO et al., 2019).

2.5.5 LSTM

Long Short-Term Memory (LSTM) units were introduced by Hochreiter and Schmidhuber in 1997 (HOCHREITER; SCHMIDHUBER, 1997) and were specifically designed to deal with the vanishing gradient problem. Differently from vanilla RNNs described in Section 2.5.4, which struggle to retain information over long sequences, LSTM cells are specifically designed to capture long-term dependencies by preserving information across many time steps (AL-SELWI et al., 2024).

The core idea behind the LSTM architecture lies in its gating mechanism, which regulates the flow of information through the cell. This mechanism is composed of three gates: the forget gate, the input gate, and the output gate, shown in Figure 10. Each gate uses a sigmoid activation function, producing values between 0 and 1 to determine how much information should pass through (SCHMIDT, 2019).

- The forget gate decides which parts of the previous cell state should be discarded, allowing the network to "forget" irrelevant or outdated information.
- The input gate determines which portions of the new input information should be added to the cell state.
- The output gate controls which parts of the internal cell state are exposed as the output hidden state at the current time step.

The forget and input gates are responsible for updating the cell state, which acts as an internal memory, carrying forward important information throughout the sequence. The output gate, meanwhile, regulates how much of the updated cell state should influence the hidden state output. Through this mechanism, LSTMs are able to maintain and regulate long-term information flow (AL-SELWI et al., 2024).

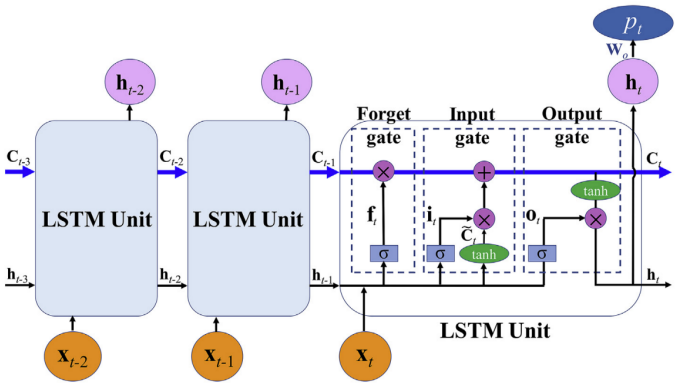


Figure 10 – Structure of an LSTM cell with input, forget, and output gates. Adapted from (WEI et al., 2021)

The LSTM unit can be mathematically described as follows. Given an input vector $\boldsymbol{X}_t$ at time step t and the previous hidden state $\boldsymbol{h}_{t-1}$, the LSTM computes the output gate $\boldsymbol{o}_t$, the input gate $\boldsymbol{i}_t$, and the forget gate $\boldsymbol{f}_t$ using the following equations:

$$\boldsymbol{o}_t = \sigma\left(\boldsymbol{X}_t \boldsymbol{W}_{xo} + \boldsymbol{h}_{t-1} \boldsymbol{W}_{ho} + \boldsymbol{b}_o\right) \tag{2.34}$$

$$\boldsymbol{i}_t = \sigma\left(\boldsymbol{X}_t \boldsymbol{W}_{xi} + \boldsymbol{h}_{t-1} \boldsymbol{W}_{hi} + \boldsymbol{b}_i\right) \tag{2.35}$$

$$\boldsymbol{f}_t = \sigma\left(\boldsymbol{X}_t \boldsymbol{W}_{xf} + \boldsymbol{h}_{t-1} \boldsymbol{W}_{hf} + \boldsymbol{b}_f\right) \tag{2.36}$$

After this computation, the LSTM unit computes the candidate memory cell $\tilde{\boldsymbol{C}}_t$, which represents the candidate information that could be added to the cell state, as:

$$\tilde{\boldsymbol{C}}_t = \tanh\left(\boldsymbol{X}_t \boldsymbol{W}_{xc} + \boldsymbol{h}_{t-1} \boldsymbol{W}_{hc} + \boldsymbol{b}_c\right) \tag{2.37}$$

After that, the LSTM introduces the previous memory content $\boldsymbol{C}_{t-1}$, which, together with the other gates, controls how much of the old memory information will be retained in the new memory state $\boldsymbol{C}_t$, which can be computed using:

$$\boldsymbol{C}_t = \boldsymbol{f}_t \otimes \boldsymbol{C}_{t-1} + \boldsymbol{i}_t \otimes \tilde{\boldsymbol{C}}_t \tag{2.38}$$

where $\otimes$ denotes the element-wise multiplication. Finally, the LSTM unit computes the hidden state $\boldsymbol{h}_t$ as:

$$\boldsymbol{h}_t = \boldsymbol{o}_t \otimes \tanh(\boldsymbol{C}_t) \tag{2.39}$$

It is important to notice the hidden state $\boldsymbol{h}_t$ is used as the output of the LSTM cell and, together with the memory state $\boldsymbol{C}_t$, is used in the computation of the next time step.

2.5.6 Attention

Attention is an important concept in deep learning that was first introduced by Bahdamau, Cho and Bengio (BAHDANAU; CHO; BENGIO, 2016) in a machine

translation problem and gained widespread popularity with the famous paper "Attention Is All You Need" proposed by Vaswani et al. (VASWANI et al., 2017), where the authors proposed the transformer architecture. Inspired by the human attention of selectively focusing on relevant information, attention mechanisms enable neural networks to assign different weights to different parts of the input, allowing the model to concentrate on the most important elements of the data. This selective focus not only improves the model's interpretability but also makes it possible to handle large amounts of information more efficiently, reducing computational overhead (NIU; ZHONG; YU, 2021).

The attention mechanism works by taking a vector of features $\boldsymbol{F}$, which are obtained after the input data $\boldsymbol{X}$ is passed through a feature extraction model to extract the feature vectors $\boldsymbol{f}_1$, to $\boldsymbol{f}_n$. This feature extractor can vary depending on the task and, in the case of time series analysis, for example, it could be an encoder composed of CNNs, MLPs, or LSTMs, that produces the latent representations $\boldsymbol{f}_1$ to $\boldsymbol{f}_n$ (BRAUWERS; FRASINCAR, 2023).

From the feature vectors $\boldsymbol{F}$ the model then extracts the key and value vectors $\boldsymbol{K}$ and $\boldsymbol{V}$, respectively. These vectors are usually obtained through a linear transformation of $\boldsymbol{F}$ using the weight matrix $\boldsymbol{W}_K$ and $\boldsymbol{W}_V$.

After that, the model introduces a query vector $\boldsymbol{Q}$, which is a task-specific vector that guides the attention mechanism to focus on the most important vectors (BRAUWERS; FRASINCAR, 2023; NIU; ZHONG; YU, 2021). From the query and key vectors, the model then computes the energy score $\boldsymbol{e}$ using a score function f:

$$\boldsymbol{e} = f\left(\boldsymbol{Q}, \boldsymbol{K}\right) \tag{2.40}$$

where f can be a variety of functions, such as the scaled dot product (VASWANI et al., 2017):

$$f\left(\boldsymbol{Q}, \boldsymbol{K}\right) = \frac{\boldsymbol{Q}^T \boldsymbol{K}}{\sqrt{d_k}} \tag{2.41}$$

The energy scores $\boldsymbol{e}$ are then mapped to the attention weights $\boldsymbol{\alpha}$ using an attention distribution function g, usually the softmax function (NIU; ZHONG; YU, 2021):

$$\boldsymbol{\alpha} = \mathrm{g}(\boldsymbol{e}) \tag{2.42}$$

Using the attention weights $\boldsymbol{\alpha}$ and the value vector $\boldsymbol{V}$, the model computes the context vector $\boldsymbol{c}$ using:

$$\boldsymbol{c} = \sum_{l=1}^{n} \alpha_l \boldsymbol{V}_l \tag{2.43}$$

where $\alpha_l \in \mathrm{R}^1$ is the weight of the l-th vector. Figure 11 shows the architecture of the attention model.

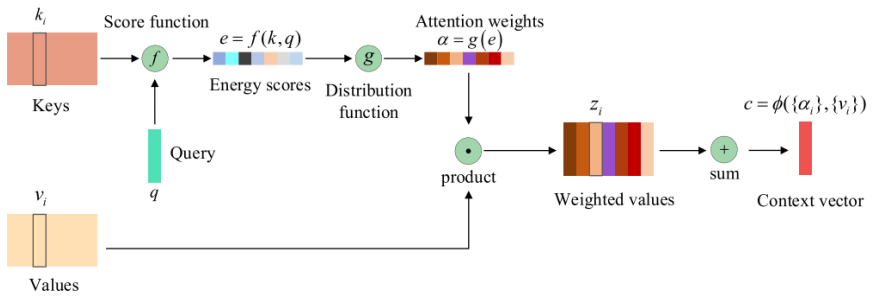


Figure 11 – Diagram of the attention mechanism showing query, key, value vectors, and the computation of the context vector. (NIU; ZHONG; YU, 2021)

2.5.7 Transformers

The transformer architecture was introduced by Vaswani et al. (VASWANI et al., 2017) and, since then, has gained widespread popularity in various fields, such as natural language processing (NLP), computer vision (CV) and time series analysis (LIN et al., 2021). It is built upon the attention mechanism, shown in Section 2.5.6, which makes it particularly effective in modeling long range dependencies in sequential data (WEN et al., 2023). A variety of transformers architectures have been proposed to improve the original transformer, as shown in Lin et al. (LIN et al., 2021). However, this section will focus on the vanilla Transformer.

The vanilla transformer architecture is a sequence to sequence model composed of an encoder and a decoder. The encoder maps an input sequence $\boldsymbol{X} = (x_1, ..., x_n)$ to a latent representation $\boldsymbol{Z} = (z_1, ..., z_n)$. The decoder then utilizes this latent space to generate an output sequence $\boldsymbol{Y} = (y_1, ..., y_m)$ one step at a time. The process is auto-regressive, meaning it uses the previously generated output to generate the next output (VASWANI et al., 2017). Figure 12 shows the architecture of the transformer model.

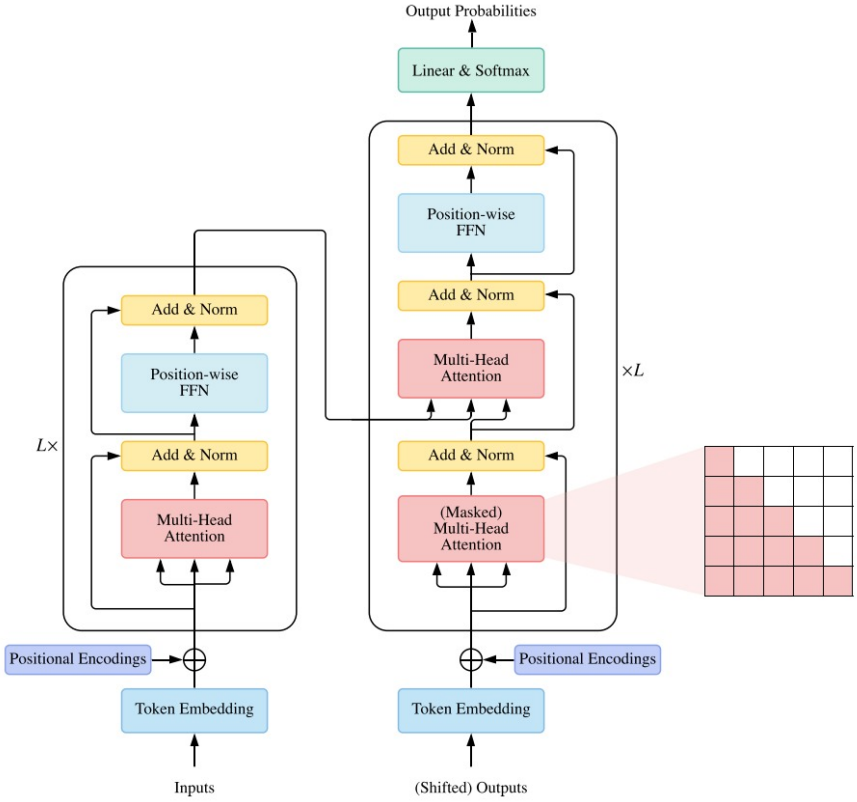


Figure 12 – Architecture of the vanilla Transformer model, with encoder and decoder components. (LIN et al., 2021)

The encoder is composed of a stack of L identical blocks. Each block is composed by two sub-layers: a multi-head attention and a feed-forward network. The multi-head attention sublayer is composed of H attention blocks, that use different projection matrices $\boldsymbol{W}_K$, $\boldsymbol{W}_V$ and $\boldsymbol{W}_Q$ to compute the key, value and query vectors, respectively. The output of each attention block is then concatenated and multiplied by a matrix $\boldsymbol{W}^O$, as shown in Equation 2.44:

$$\text{MultiHead}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = (\text{head}_1; ...; \text{head}_H) \cdot \boldsymbol{W}^O \tag{2.44}$$

where $\text{head}_i = \text{Attention}(\boldsymbol{Q}\boldsymbol{W}_i^Q, \boldsymbol{K}\boldsymbol{W}_i^K, \boldsymbol{V}\boldsymbol{W}_i^V)$ is the i-th attention head, and Attention is the attention function defined in Section 2.5.6.

A residual connection and layer normalization follow the multi-head attention, as shown in Equation 2.45:

$$\boldsymbol{H}' = \text{LayerNorm}(\text{MultiHead}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) + \boldsymbol{X}) \tag{2.45}$$

where $\boldsymbol{X}$ is the input to the multi-head attention sublayer and LayerNorm is the layer normalization function. The output $\boldsymbol{H}'$ is then processed by a position-wise feed-forward network with two fully connected layers: 2.46:

$$\text{FFN}(\boldsymbol{H}') = \text{ReLU}(\boldsymbol{H}'\boldsymbol{W}_1 + \boldsymbol{b}_1)\boldsymbol{W}_2 + \boldsymbol{b}_2 \tag{2.46}$$

The decoder has a similar architecture to the encoder. It has L stacked blocks, but each block is composed of three sub-layers: a masked multi-head attention layer, a multi-head attention layer, and a feed-forward network. The masked multi-head attention layer works similarly to the multi-head attention, but the self-attention is masked to prevent the model from attending to future tokens in the sequence, ensuring that the prediction for a given token only depends on the previous tokens (LIN et al., 2021). Another difference is in the second multi-head attention layer, which uses the output of the encoder $\boldsymbol{Z}$ to project the key and value vectors of the attention.

Finally, one important point to highlight is the positional encoding layer. In vanilla transformers, the model does not have access to the order of the sequence. To deal with this, Vaswani et al. (VASWANI et al., 2017) propose adding a positional encoding to the input. They do so using the sine and cosine functions:

$$PE_{(pos,2i)} = \text{sen}\left(\frac{pos}{10000^{2i/d_{model}}}\right) \tag{2.47}$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right) \tag{2.48}$$

where i is the dimension, pos is the token position and d_{model} is the model dimension.

2.6 Reconstruction-Based Models

Reconstruction-based models are a class of machine learning models that are designed to learn the intrinsic correlations in data by trying to reconstruct the original input signal. Differently from forecasting-based approaches, which attempt to predict future values, reconstruction-based models focus on accurately recreating the current input. This approach makes them great for anomaly detection, often outperforming forecasting-based models because they have access to the current time series data, which helps them to be more stable to rapidly changing time series. However, this advantage comes with a trade-off because they introduce a delay in the detection of anomalies as they rely on reconstruction error analysis. (DARBAN et al., 2024).

The main idea behind these networks is to train them to accurately reconstruct normal data, which is presumed to be the majority of the available data. This is usually done by minimizing a reconstruction loss, such as the Mean Squared Error (MSE), during the training phase.

During the test phase, the model encounters anomalous data, which it has not been trained to reconstruct, causing the reconstruction error to increase greatly compared to the error of normal data, as the model fails to reconstruct abnormal patterns. This way, an anomaly can be identified by thresholding the reconstruction error. Inputs with low reconstruction error will be classified as normal, and inputs with reconstruction error greater than the threshold will be classified as anomalous.

There are a variety of reconstruction-based models. In Lachekhab et al. (LACHEKHAB et al., 2024), the authors proposed an autoencoder composed of multiple LSTM cells to identify anomalies in electrical motors vibrations. The model was trained using the MSE loss function to reconstruct normal vibration patterns, and anomalies were detected by applying a threshold to the reconstruction error.

In Xie, Xu and Jiang (XIE; XU; JIANG, 2023), the authors developed a model that combines convolutional layers, LSTM layers and a variational autoencoder to detect anomalies in multivariate time series. They computed correlation matrices between the variables to use as an input for the model and used a loss function that combines the MSE loss to the Kullback-Leibler divergence to train the model, achieving state-of-the-art results in three different datasets.

Another example is the model proposed in Tuli, Casale and Jennings (TULI; CASALE; JENNINGS, 2022) where authors utilized the transformer architecture to detect anomalies. The model works on two phases: in the first phase the model tries to reconstruct the input sequence, obtaining a reconstruction error. From this, the model computes the focus score that highlight areas with bigger reconstruction errors. In the second phase, the focus score is used by the attention mechanism to focus on areas where the reconstruction error was bigger, allowing the model to concentrate on these high-error regions and attempt a refined reconstruction. This approach allowed the authors to outperform state-of-the-art models on six different datasets.

2.7 Railway Components

The railway track consists of different interdependent components that are divided into two categories: the superstructure and the substructure. The former comprises the rails, fastenings, and sleepers, while the latter consists of the ballast, sub-ballast, and subgrade (KAEWUNRUEN[1]; REMENNIKOV[1], 2008), as shown in Figure 13. These two structures are separated by the sleeper-ballast interface and are essential for ensuring a safe and cost-effective transportation system capable of guiding vehicles and transmitting loads to the subgrade (ATTOH-OKINE, 2017).

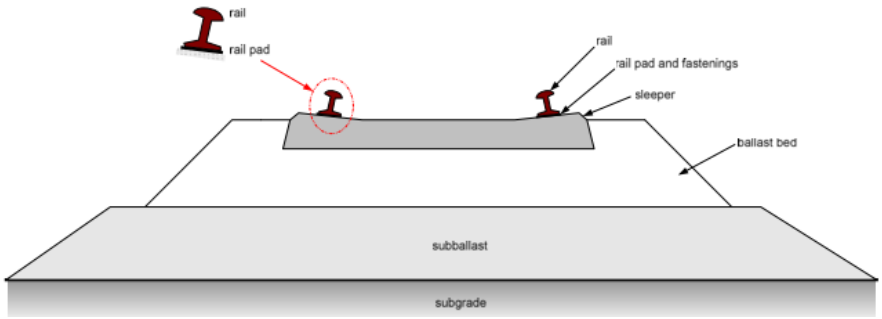


Figure 13 – Main railway track components, including superstructure and substructure. (KAEWUNRUEN[1]; REMENNIKOV[1], 2008)

2.8 Track Irregularities Standards

Railway track irregularities are deviations from the ideal track geometry that can affect the safety and comfort of train operations (SANSINENA; RODRíGUEZ-ARANA; ARRIZABALAGA, 2025). So, to ensure safe and efficient railway operations, a continuous assessment of track condition is necessary to support planning of maintenance actions. For this reason, different countries have established standards that quantify the characteristics of the track and their acceptable limits. In subsections 2.8.1 and 2.8.2, two of these standards will be presented: the Federal Railroad Administration (FRA) standard, used in the United States, and the Brazilian standard, used in Brazil.

2.8.1 FRA Standard

Railway track irregularities are normally modeled as being random and, for this reason, are typically described by a PSD function, see Section 2.2.3. Countries like USA, China, Britain and Germany adopted this approach to describe track irregularities (PIRES et al., 2021; BERAWI, 2013). The PSD function parameters were estimated from measured data and, therefore, represent the average irregularity observed in each respective railway network.

FRA classifies tracks into nine categories based on their condition, assigning a PSD function to each category. Categories 1 to 6 are designed for ordinary tracks, while categories 7 to 9 are intended for high-speed tracks. Due to equipment limitations, the FRA standard can only describe irregularities with wavelengths between 1.524 m to 304.8 m (BERAWI, 2013).

The PSD formulas are given by Equations 2.49, 2.50 and 2.51:

$$S_{av}(\Omega) = \frac{k \cdot A_v \cdot \Omega_c^2}{\Omega^2 \cdot (\Omega^2 + \Omega_c^2)} \tag{2.49}$$

$$S_{al}(\lambda) = \frac{k \cdot A_a \cdot \Omega_c^2}{\Omega^2 \cdot (\Omega^2 + \Omega_c^2)} \tag{2.50}$$

$$S_{gauge/cl}(\lambda) = \frac{4 \cdot k \cdot A_v \cdot \Omega_c^2}{(\Omega^2 + \Omega_c^2) \cdot (\Omega^2 + \Omega_s^2)} \tag{2.51}$$

where $S_{av}(\Omega)$ is the vertical alignment PSD, $S_{al}(\Omega)$ is the lateral alignment PSD, and $S_{gauge/cl}(\Omega)$ is the gauge or the superelevation PSD, all expressed in $[cm^2/(rad/m)]$. The variable Ω is the spatial wavenumber, while Ω_c and Ω_s are the critical wavenumbers, all in $[rad/m]$. The variable k is a constant, and A_v and A_a are the roughness coefficients related to the line grade, expressed in $[cm^2 \cdot rad/m]$. Table 1 summarizes the parameters for each track class.

Class	Max velocity (km/h)		Parameters			
	Freight	Passengers	A_v (cm^2 rad/m)	A_a (cm^2 rad/m)	Ω_c^2 (rad/m)	Ω_s^2 (rad/m)
1	16	24	1.2107	3.3634	0.6046	0.8245
2	40	48	1.0181	1.2107	0.9308	0.8245
3	64	97	0.6816	0.4128	0.8520	0.8245
4	97	129	0.5376	0.3027	1.1312	0.8245
5	129	145	0.2095	0.0762	0.8209	0.8245
6	177	177	0.0339	0.0339	0.4380	0.8245

Table 1 – Summary of the parameters for each track class. (PIRES et al., 2021)

One important point to highlight is that track condition in USA have very similar conditions to those in Brazil. For this reason, the FRA standard can be used to describe track irregularities in Brazil (PIRES et al., 2021).

2.8.2 Brazilian Standard

The Brazilian Standard ABNT NBR 16387/2020 (ABNT, 2020) specifies the geometrical limits for various railway track parameters, as presented in Table 2. It is important to note that these limits are related to the track irregularity profile but are not identical to it. To obtain the corresponding value for each geometry parameter, a specific function must be applied to the irregularity profile, converting it into a value that can be directly compared to the standard's limits. This procedure will later be used to label the track profiles, with irregularities exceeding the limits classified as defects.

Description of track geometry parameter	**Gauge**	**(0–15) km/h**	**(16–40) km/h**	**(41–64) km/h**	**(65–95) km/h**	**(96–128) km/h**
Gauge						
Open gauge limit (Static gauge)	Meter	1035 mm	1032 mm	1032 mm	1025 mm	1013 mm
Closed gauge limit (Static gauge)	Meter	987 mm	987 mm	987 mm	987 mm	987 mm
Rapid gauge variation in 5 m (Static gauge)	Meter	34 mm	31 mm	23 mm	18 mm	13 mm
Crosslevel						
Crosslevel variation every 2 m (Tangent)	Meter	47 mm	34 mm	21 mm	7 mm	6 mm
Crosslevel variation every 2 m (Curve)	Meter	18 mm	14 mm	11 mm	5 mm	3 mm
Crosslevel variation every 10 m	Meter	53 mm	37 mm	33 mm	31 mm	30 mm
Alignment						
Curve misalignment in 10 m: Maximum horizontal offset between adjacent points measured every 2.5 m at the mid-chord of 10 m	Meter	31 mm	24 mm	18 mm	13 mm	8 mm
Tangent alignment defect: Maximum horizontal offset between adjacent points measured every 2.5 m at the mid-chord of 10 m	Meter	31 mm	24 mm	18 mm	13 mm	8 mm
Curvature						
Maximum superelevation in curve	Meter	100 mm	100 mm	100 mm	100 mm	100 mm

Table 2 – Limit values for the track geometry parameters. (ABNT, 2020)

2.9 How to Generate Track Irregularities from PSDs

As stated in Section 2.8.1, track irregularities are modeled as random processes described by a PSD function. The PSDs, however, does not describe the track irregularity profile directly, but rather the distribution of the energy of the signal over the spatial frequency Ω. To obtain the track irregularity, it is used an approach called the spectral representation method (HU; SCHIEHLEN, 1997), for more details see Appendix D. The idea behind this method is to split the PSD into N grids that are spaced by $\Delta\Omega$ and, each grid is then represented by a cosine wave with frequency Ω_i and the track irregularity profile is then obtained by the sum:

$$r(\chi) = \sum_{i=1}^{N} A_i \cos(\Omega_i \chi + \phi_i) \tag{2.52}$$

where χ is the spatial coordinate, $A_i = \sqrt{2S(\Omega_i)\Delta\Omega}$ is the amplitude of the i-th cosine wave, Ω_i is the spatial frequency, $\phi_i \sim \mathcal{U}(0, 2\pi)$ is a pseudo-random phase, and $S(\Omega_i)$ is the PSD function evaluated at Ω_i.

2.10 Assessing Railway Track Condition

Maintaining railway tracks in good condition is crucial to ensure safe and comfortable operations of trains (TSUNASHIMA, 2019). This maintenance is achieved through continuous monitoring of track conditions and the detection of irregularities. There are now two main methods to assess track quality: using track geometry cars (TGC) to directly measure the physical geometry of the track, and monitoring the dynamic response of the train using instrumented railway vehicles (IRV) equipped with onboard sensors capable of measuring the dynamics of the system (PIRES et al., 2024).

TGCs are equipped with sophisticated systems that directly measure the geometry of the track, that can be later compared to regulatory standards limits to identify parts of the track that need maintenance. However, this method has several drawbacks (GHIASI et al., 2025):

- **Operational disruption**: the operation of the railway track needs to be disrupted during the TGC inspection, which can halt operations for several hours depending on the length of the track being measured;
- **Operational high cost**: the sophisticated equipment and logistics make TGC operation costly, which limits their frequent use;
- **Low inspection frequency**: normally, the TGC measures the condition of the track monthly, due to its high cost, which is a problem if a fault appears between inspections.

To overcome the limitations in TGCs, an alternative approach was developed using IRVs equipped with sensors that measure the vehicle's dynamic response due to track excitations. The underlying principle behind IRVs is that the vehicle's vibrations are deeply correlated to track excitations, and an irregularity in the track will cause an anomalous dynamic response from the vehicle, which can be identified in the sensors' readings.

Using IRVs instead of the TGCs has several advantages, such as (ONO et al., 2023):

- **No disruptions**: the operation is not halted during the measurements, since data can be collected during normal train services, which reduces the costs with logistics;
- **Real time data collection**: data is collected in real time, which speeds up possible defect detections;
- **High frequency of inspections**: Since the measurements are taken directly from

in-service trains, IRVs enable near-continuous monitoring of track conditions, greatly increasing the inspection frequency.

Despite that, it is important to highlight that data collection using IRVs also comes with some drawbacks, such as (GHIASI et al., 2025):

- High amount of data: the amount of data collected during IRV operation can be huge. This data needs to be carefully processed to obtain good quality data;
- Domain problem: different operations conditions, such as the wagon mass or velocity, can affect the dynamic response of the vehicle, so data collected from a one part of the track can be inherently different from another part;
- Difficult Correlation: since the measurements are done indirectly, they need to be correlated to the track condition, which can be a complex task that often needs the use of deep learning models;
- Data imbalance: since fault measurements are much more uncommon than normal measurements, the machine learning model needs to deal with class imbalance.

2.11 Effect of the Velocity on Train Dynamics

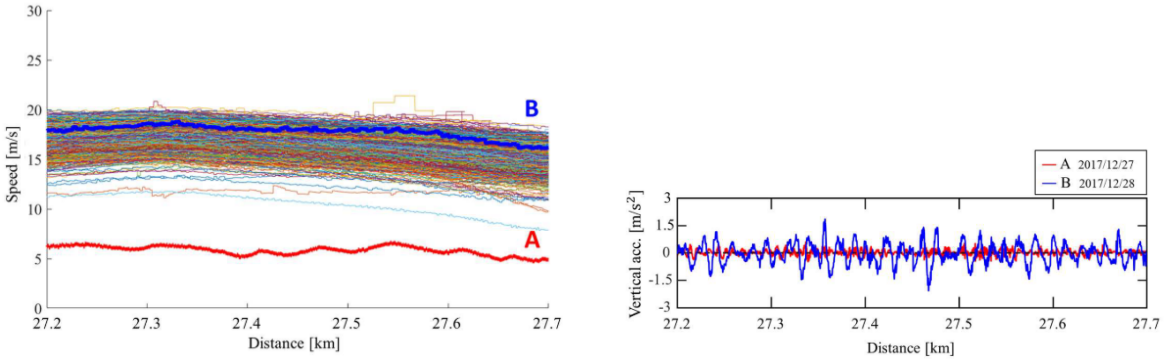
Train vibration measurements are strongly correlated to the velocity at which these measurements took place. As the wagon moves along the track, changes in velocity can alter the vibration responses measured by onboard sensors. Figure 14 shows the difference in measurement found by (ONO et al., 2023). In Figure 14(a), the authors highlight two distinct velocity profiles labeled A and B, where the speed ratio between B and A is approximately 3. This difference in speed causes a significant change in the measured signal, as shown in Figure 14(b), where the velocity profile B resulted in higher acceleration measurements than A despite them traversing the same part of the track. At lower speeds, the dynamic response of the train to the track excitations tends to yield lower acceleration values, while, higher velocities tend to generate higher values of acceleration measured.

To address this velocity-induced distortion, the authors propose two different correction methods. The first method involves using the Mahalanobis distance (MAESS-CHALCK; JOUAN-RIMBAUD; MASSART, 2000) to distinguish outliers from normal data. After that, the authors fitted a linear regression to normal data to predict the expected acceleration given the speed. The measurements are then normalized by dividing them by their predicted values.

In the second method, the authors employed a Gaussian process regression (WILLIAMS; RASMUSSEN, 1995) to model the behavior of acceleration and speed,

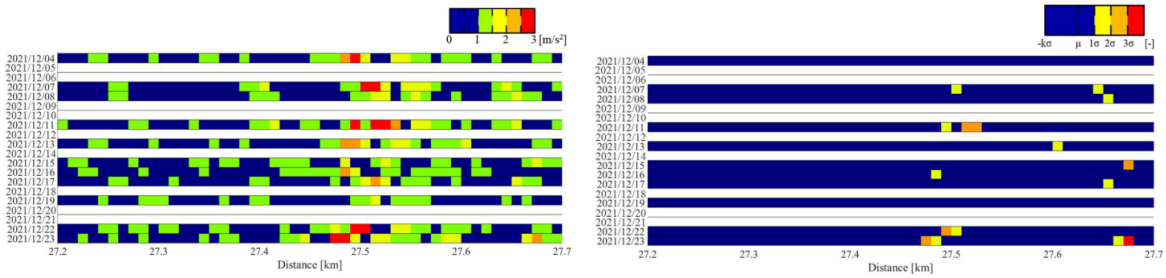
obtaining a regression that mapped the velocity to the expected acceleration. Similarly, they normalized the measurements by their respective predictions.

Both methods proved to be effective in mitigating the velocity effect, which reduced the number of false positives in anomaly detection, as shown in Figure 15.



(a) (b)

Figure 14 – Example of how velocity differences affect acceleration measurements on the same track segment. (***a***) shows the two velocity profiles, A and B. (***b***) illustrates the corresponding acceleration measurements for each profile. Adapted from (ONO et al., 2023).



(a) (b)

Figure 15 – Reduction of false positives in anomaly detection after velocity correction. (***a***) shows the heatmap of the acceleration measurements before correction, while (***b***) shows the heatmap after correction. Adapted from (ONO et al., 2023).

Another example demonstrating the impact of the velocity on measured acceleration is presented in Balouchi, Bevan and Formston (BALOUCHI; BEVAN; FORMSTON, 2021). In this work, the authors utilized the VAMPIRE vehicle dynamics simulation software to model a vehicle operating in a measured track geometry at a variety of speeds. The simulated results were then compared to the actual car body vibration measurements recorded over the same section of the track. The results are shown in Figure 16, where the continuous colored lines are the simulation results, blue circles represent the maximum acceleration found in the simulation, and the dashed lines are the car body measurements.

The authors fitted two models to the maximum simulated accelerations: a linear

regression (green line) and an exponential model (purple curve), as shown in Figure 16. Based on the higher R^2 value, they selected the exponential model as the more suitable fit. They further argue that, at lower speeds, the acceleration response does not provide sufficient confidence to distinguish whether the track quality is good or poor. Therefore, a normalization approach, similar to that proposed by One et al. (ONO et al., 2023), is required.

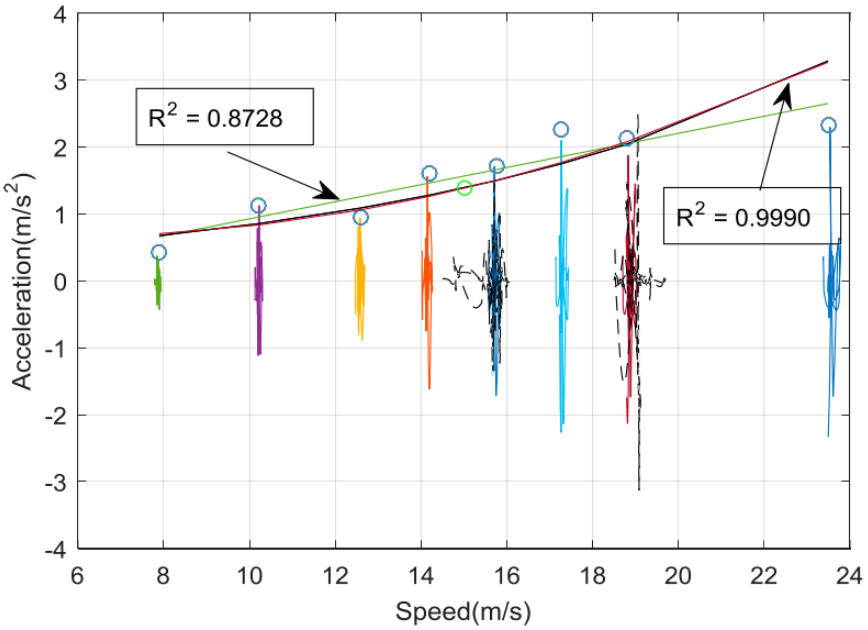


Figure 16 – Effect of train velocity on measured acceleration compared with simulation results and the linear and exponential model fitted to the maximum acceleration values with their corresponding R^2 values (BALOUCHI; BEVAN; FORMSTON, 2021).

In the two papers above, the authors show the importance of correcting the acceleration measurements acquired at different velocities, as variations in speed alter the dynamic response of the train. However, their correction only considers the effect of the velocity and not the effect of the mass, as these studies were performed in a passenger car. In this type of vehicle, the mass doesn't vary significantly, as it carries passengers which are way lighter compared to the raw materials transported by heavy haul wagons. Section 2.12 will show that the mass of the train might also affect the dynamics of the system.

2.12 Effect of Mass on Train Dynamics

When analyzing the data collected by an IRV in a Brazilian railway, described in detail in Pires et al. (PIRES et al., 2024), it was noticed that the mass of the wagon influences the relationship between acceleration growth and velocity. This phenomenon is illustrated in Figure 17, which shows the 99th percentile of the absolute maximum vertical

acceleration as a function of the velocity class. Each velocity class corresponds to data points that have a velocity between an 1 m/s interval. For example, the velocity class $7 < v < 8$ corresponds to all data points whose velocity is between 7 m/s and 8 m/s.

In the Figure, data points are colored according to the wagon mass class. The unloaded class corresponds to an empty wagon, i.e, that is not transporting any material, the loaded class corresponds to a wagon fully loaded with iron ore, and the 1/4 to 3/4 classes correspond to wagons that were either partially loaded or carrying lighter materials such as coal. One point to highlight, is that it was used the 99th percentile instead of the maximum value because the dataset consists of real-world measurements, which are prone to noise and outliers.

As shown in the Figure, different wagon masses produce distinct acceleration growth behavior: the unloaded wagon exhibits the lowest acceleration growth, the 1/4 loaded wagon shows a slightly higher growth, and the 2/4, 3/4, and fully loaded wagons display similar growth patterns.

These observations highlight the need to further investigate the correlation between wagon mass and velocity behavior for freight operations.

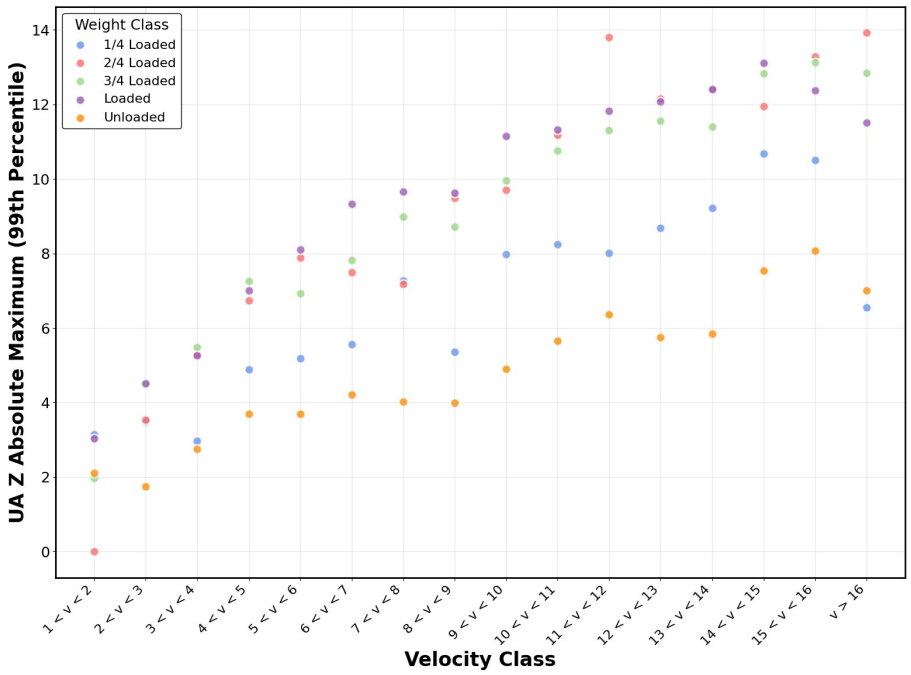


Figure 17 – Effect of wagon mass on the relationship between acceleration and velocity for freight trains. The data points represent the 99th percentile of the absolute maximum vertical acceleration for each velocity class, colored by wagon mass class.

TEM QUE VOLTAR NESSA SECAO E ADICIONAR REFERENCIAS.

2.13 Normalization of Signals for Different Operational Conditions

As discussed in Sections 2.11 and 2.12, the different operational conditions cause variations in the dynamic responses measured by the IRV, like the increase of acceleration amplitude and the drift in the frequency spectrum. These variations in the measured signals can be problematic specially for machine learning models. For example, a model trained on data from a wagon moving near 65 km/h might not be able to identify defects in the track when feeded with data from a wagon moving at 20 km/h. For this reason, it is important to remove these variations caused by different operational conditions, which will allow the model to become more robust.

There are an extensive literature on how to normalize signals for different operational conditions. In the next paragraphs, some of the methods are presented.

In Urbanek et al. (URBANEK et al., 2017), the authors first defined what they call the generalized angular deterministic (GAD) signal. They considered a shaft with angular position $\phi(t)$, angular velocity $\dot{\phi}(t)$ and subject to a load $l(t)$. They assume that the change in operational conditions vary slowly and smoothly. Therefore, for a signal $x(t)$, sampled in equal angular increments $x(\varphi(t)) = x'(\phi)$, is defined as GAD with given period Φ when the function composition $(z \circ \varphi)(t)$ satisfies the following criteria:

$$z(\varphi) = z(\varphi + \Phi) \tag{2.53}$$

Using this definition, the authors proposed the following normalization method:

$$z(\varphi) = \frac{x'(\varphi)}{g'_f(\varphi)} \tag{2.54}$$

where $g'_f(\varphi)$ is the low-pass filtered version of $g(\varphi)$, with the later being the resampled version of $g(t) = |x(t) + i\mathcal{H}[x(t)]|$. To determine the cutoff frequency of the low-pass filter the authors the authors used the following formula:

$$f = \max\left(|\Delta_{\dot{\varphi}}|, |\Delta_l|\right) \tag{2.55}$$

where $\Delta_{\dot{\varphi}}$ and Δ_l are given by:

$$\Delta_{\dot{\varphi}} = \max\left(\frac{d\dot{\varphi}(t)}{dt} \cdot \frac{1}{\dot{\varphi}(t)}\right) \tag{2.56}$$

$$\Delta_l = \max\left(\frac{dl(t)}{dt} \cdot \frac{1}{l(t)}\right) \tag{2.57}$$

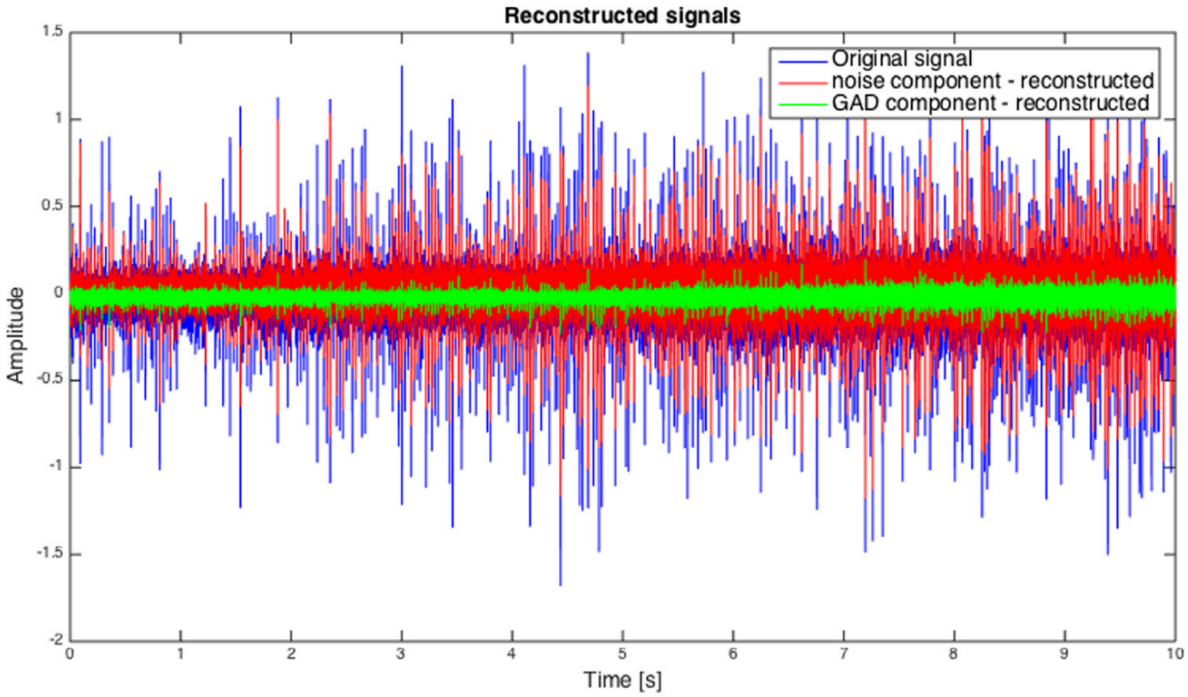
Assuming that the load and speed variations are the only source of non-stationarity, the resulting signal $z(\varphi)$ is purely periodic. By also assuming that the STFT (see Section 2.3.1) of $z(\varphi)$ is the same throughout the entire signal, the authors proposed an extension of the Discrete Randon Separation (DRS) (ANTONI, 2005) method using the formula:

$$G(\Theta) = \frac{\langle Z(\Theta, \psi) Z^*(\Theta, \psi + \Phi) \rangle_\psi}{\langle |Z(\Theta, \psi)|^2 \rangle_\psi} \tag{2.58}$$

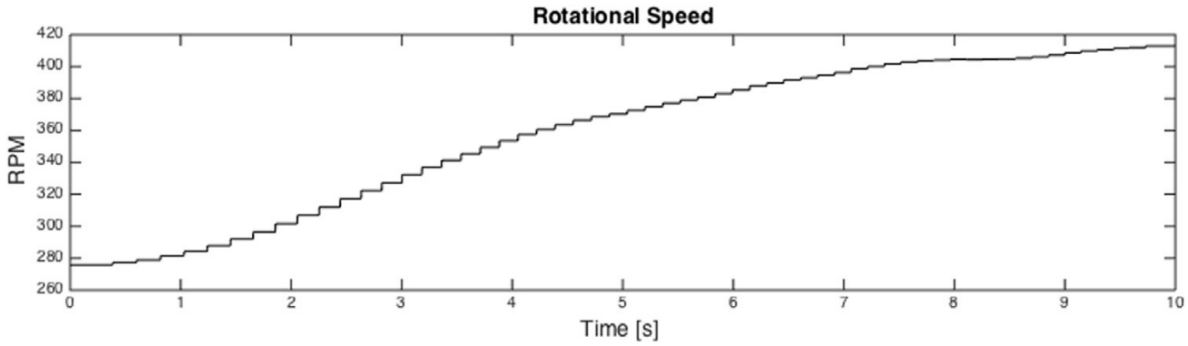
where $Z(\Theta, \psi)$ is the STFT of $z(\varphi)$, $\langle \cdot \rangle_\psi$ is the average over ψ, and $G(\Theta)$ is the transfer function. The full procedure proposed by the authors is then:

1. Resample the signal $x(t)$ to obtain $x'(\varphi)$;
2. Compute the GAD signal $z(\varphi)$;
3. Compute the transfer function $G(\Theta)$;
4. Convolve the transfer function $G(\Theta)$ with $x'(\varphi)$;
5. Return the signal to the time domain, giving the deterministica pat of the GAD. The residual is the random part.

The authors then tested this method in 3 different case studies: for a simulated signal, for a signal acquired in a laboratory experiment and for a real-world signal acquired from a industrial printing machine. For the sake of simplicity, it is shown only the results for the printing machine. Figure



(a)



(b)

Figure 18

2.14 Assessing Track Irregularities from Acceleration Data

In Sansinena, Rodríguez-Arana and Arrizabalaga (SANSINENA; RODRíGUEZ-ARANA; ARRIZABALAGA, 2025) authors presented a review of research papers that proposed methods for estimating track irregularities using acceleration data. They divided the methods into three main categories:

- Model-driven methods;
- Data-driven methods;
- Hybrid methods.

In model-driven methods, a mathematical representation of the dynamic system is developed and validated on real data. Once the model accurately represents the system dynamics, the problem is inverted to estimate track irregularities from the measured acceleration data. The main advantage in model-driven methods is that they don't require a large amount of data to be designed, as they're based on expert knowledge.

For instance, in De Rosa, Alfi and Bruni (De Rosa; ALFI; BRUNI, 2019) the authors used a 3D multibody model that was previously validated to measure the yaw and roll motions of the vehicle. Based on simulation data, they applied three reconstruction techniques, the Kalman filter, the Unknown Input Observer (UIO) and the Frequency Response Function (FRF), to reconstruct the lateral irregularities and the cross alignment. The performance of these methods were compared using two evaluation metrics J_y and J_ρ, defined by Equations 2.59 and 2.60:

$$J_y = \frac{\text{rms}(y_{est} - y)}{\text{rms}(y)} \tag{2.59}$$

$$J_\rho = \frac{\text{rms}(\rho_{est} - \rho)}{\text{rms}(\rho)} \tag{2.60}$$

where y_{est} and ρ_{est} are the estimated lateral and cross alignment irregularities, y and ρ are the original values and rms is the root mean square function. The results for these two metrics are shown in Table 3, where values closer to 0 indicate a better performance and FD corresponds to the FRF function.

	FD	UIO	KF
$\mathbf{J_y}$	0.325	0.291	0.363
$\mathbf{J_p}$	0.122	0.549	0.124

Table 3 – Comparison of the accuracy indices.

From these results, the authors decided to use the FRF method on real measured data, which included lateral and vertical accelerations recorded at the axle-boxes, bogies, and car body. However, they did not report any quantitative metrics for the reconstruction of real irregularities, presenting only the results shown in Figure 19. The authors argue that although some similarity between the reconstructed and the original irregularities can be seen, the method did not produce visually satisfactory results. They justify that this deviation is caused by non-linear effects they didn't consider, but these deviations can occur because of the lack of robustness of the method to real-world noise in data, which is a common problem when dealing with simpler reconstruction methods.

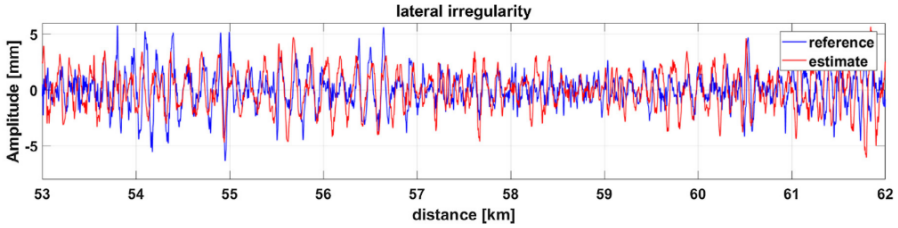


Figure 19 – Reconstruction of lateral irregularities using the Frequency Response Function method (De Rosa; ALFI; BRUNI, 2019).

Data-driven methods, on the other hand, do not consider a mathematical model of the system, relying only on data processing to reconstruct or identify the irregularities. These methods typically utilize traditional machine learning algorithms or, more recently, deep learning architectures. Such approaches are generally more robust when dealing with real-world data, as they can learn from noisy and complex signals. This robustness, however, comes with a price, as they need a larger amount of data to be trained, especially in the case of deep learning (SANSINENA; RODRíGUEZ-ARANA; ARRIZABALAGA, 2025).

In Tsunashima and Yagura (TSUNASHIMA; YAGURA, 2024) the authors propose a method of estimating railway track irregularities from car body vibration data. They first created a multibody simulation in SIMPACK. After that, they generated 12 different track irregularities, ranging from a good condition to a degraded condition, using FRA PSD formula and converted the generated profiles into 10 meters long track irregularity sections using the 10 m-chord versine method. This transformation is shown in Figure 20.

Using these irregularity profiles, they ran simulations over a 1000 meters track at speeds in the range of 40 to 80 km/h, varying the velocity in a 10 km/h interval. For each combination of track profile and speed, they collected the vertical and lateral acceleration as well as the roll rate of the carbody. Similarly to the irregularity profiles, a downsample was also applied to the data taking the maximum vibration measured in the 10 meters section, as shown in Figure 20. From this process, they obtained a dataset that correlates the maximum vibration of the carbody to the track irregularities for varying speeds and then applied a Gaussian Process Regressor (GPR) that was able to predict the irregularity from the vibration for a certain vehicle speed, as illustrated in Figure 21.

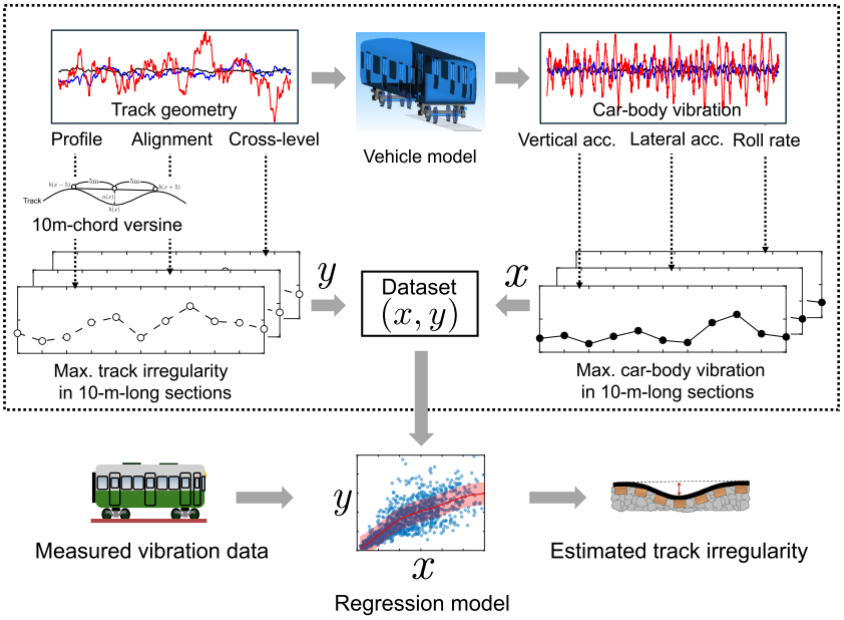


Figure 20 – Data processing pipeline for the creation of the simulated dataset (TSUNASHIMA; YAGURA, 2024).

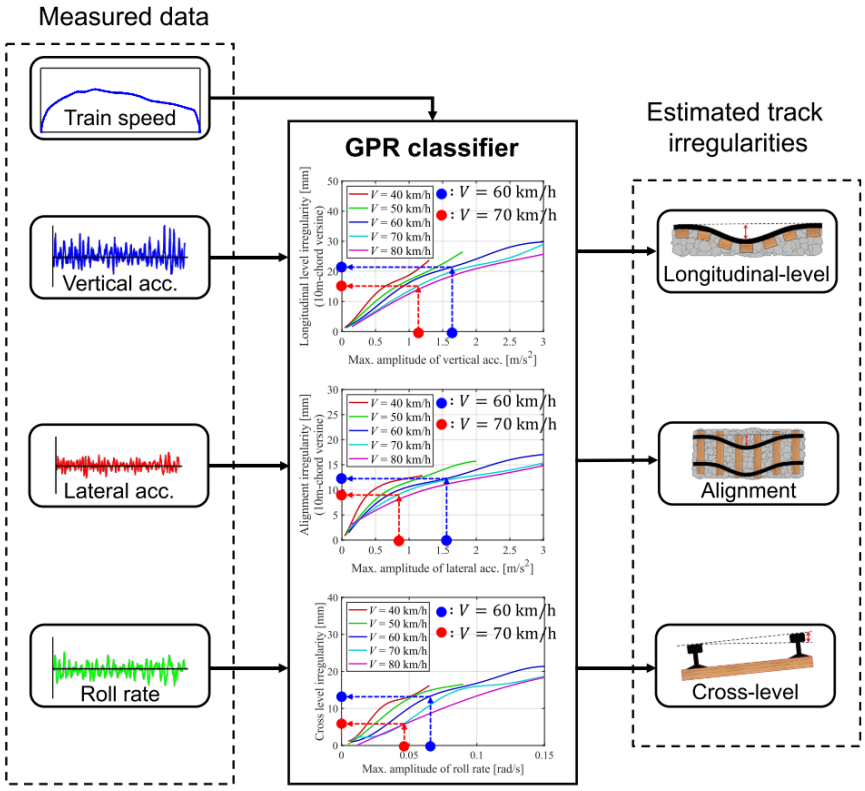


Figure 21 – Track irregularity prediction using Gaussian Process Regression for different vehicle speeds (TSUNASHIMA; YAGURA, 2024).

The authors then validated their model using real data collected with an onboard sensing device equipped with a triaxial accelerometer to measure carbody acceleration, a rate gyro to capture the carbody roll rate, and a GNSS receiver to determine vehicle

position. To ensure consistency, they applied the same downsample procedure to the real data and averaged the speed in the 10 m section to be used as an input for the regression model. The regression result for one part of the track is shown in Figure 22 and is presented with a 1σ confidence interval. However, they didn't provide any quantitative metrics.

Tsunashima and Yagura argue that for the majority of the sections, the GPR predicts the correct value within the 1σ confidence interval. However, for some of the sections, the predicted value significantly deviated from the actual measurement. They argue that there is a factor other than the track irregularity that influences the dynamics of the system and superimposes on the carbody's vibration. This difference might also occur because of the low complexity model used that might not have been able to learn the correlations between the variables very well.

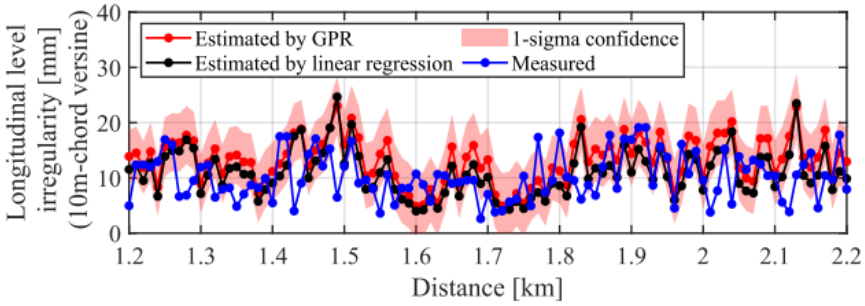


Figure 22 – Regression results comparing predicted and actual track irregularities from real measurement data. (TSUNASHIMA; YAGURA, 2024)

To address this model issue, Hao et al. (HAO et al., 2023) proposes a deep learning approach to reconstruct the track irregularity from vertical and lateral carbody acceleration measurements. The authors used a dataset composed of acceleration measurements collected in three high-speed Chinese railway lines. In the dataset, they also had access to the measured geometry of the track. Since they aimed to reconstruct irregularities with wavelengths between 3 m and 120 m, they applied a band-pass filter to remove components with wavelengths outside this range. They also decided to scale the data using a min-max scaler, although the authors are not clear if they do that to the entire dataset or if they apply it to each time window individually. Finally, since their data is measured with velocities between 40 and 310 km/h, they decided to give the vehicle speed as an input for the model.

They proposed a model that combines an attention mechanism with a 1D CNN layer and a GRU layer, allowing it to capture both spatial and temporal dependencies in the data. The model architecture is shown in Figure 23. It receives vertical and lateral carbody acceleration, along with velocity, as inputs. The output was the vertical profile, that have wavelengths between 3 and 120 m, and the long-wave vertical profile, with wavelengths between 42 and 120 m. They trained and tested the proposed model using two

different lines: the NHHSR and the BGHSR, and compared it to two simpler versions of the model. The proposed model was trained and tested on two different railway lines, the NHHSR and the BGHSR, and its performance was compared with two simpler variants of the model. The results for the three models across the two lines are shown visually in Figure 24, while the evaluation metrics are summarized in Table 4, with the best values highlighted in bold. The metrics used were the Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Theil Inequality Coefficient (TIC), and the Pearson Correlation Coefficient (PCC), defined as:

$$MAE = \frac{1}{M}\sum_{k=1}^{M}\left|y_k - \hat{y}_k\right|. \tag{2.61}$$

$$RMSE = \sqrt{\frac{1}{M}\sum_{k=1}^{M}\left(y_k - \hat{y}_k\right)^2}. \tag{2.62}$$

$$TIC = \frac{\sqrt{\frac{1}{M}\sum_{k=1}^{M}\left(y_k - \hat{y}_k\right)^2}}{\sqrt{\frac{1}{M}\sum_{k=1}^{M} y_k^2} + \sqrt{\frac{1}{M}\sum_{k=1}^{M} \hat{y}_k^2}}. \tag{2.63}$$

$$PCC = \frac{\sum_{k=1}^{M}\left(\hat{y}_k - \bar{\hat{y}}\right)\left(y_k - \bar{y}\right)}{\sqrt{\sum_{k=1}^{M}\left(\hat{y}_k - \bar{\hat{y}}\right)^2}\cdot\sqrt{\sum_{k=1}^{M}\left(y_k - \bar{y}\right)^2}}, \quad \left(\bar{y} = \frac{1}{M}\sum_{k=1}^{M} y_k,\ \bar{\hat{y}} = \frac{1}{M}\sum_{k=1}^{M}\hat{y}_k\right). \tag{2.64}$$

where y is the actual value, $\hat{y}$ is the predicted value, M is the number of samples, and $\bar{y}$ and $\bar{\hat{y}}$ are the mean values of y and $\hat{y}$, respectively.

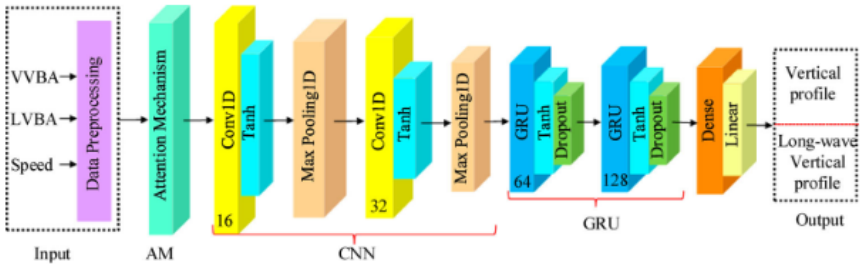


Figure 23 – Model architecture utilized to reconstruct the track irregularity (HAO et al., 2023).

HSR line	Model	Output	MAE	RMSE	TIC	PCC
NHHSR	GRU	Long-wave vertical profile	0.4995	0.6456	0.3801	0.7292
		Vertical profile	0.2720	0.3558	0.3834	0.7254
	CNN–GRU	Long-wave vertical profile	0.4951	0.6393	0.3728	0.7392
		Vertical profile	0.2688	0.3521	0.3877	0.7293
	AM–CNN–GRU	Long-wave vertical profile	**0.4808**	**0.6205**	**0.3653**	**0.7513**
		Vertical profile	**0.2652**	**0.3459**	**0.3710**	**0.7425**
BGHSR	GRU	Long-wave vertical profile	0.5048	0.6425	0.3303	0.7879
		Vertical profile	0.2934	0.3762	0.3442	0.7816
	CNN–GRU	Long-wave vertical profile	0.5020	0.6360	0.3242	0.7932
		Vertical profile	0.2905	0.3717	0.3308	0.7919
	AM–CNN–GRU	Long-wave vertical profile	**0.5001**	**0.6355**	**0.3244**	**0.7935**
		Vertical profile	**0.2867**	**0.3668**	**0.3236**	**0.7975**

Table 4 – Evaluation metrics for different models and different lines (HAO et al., 2023).

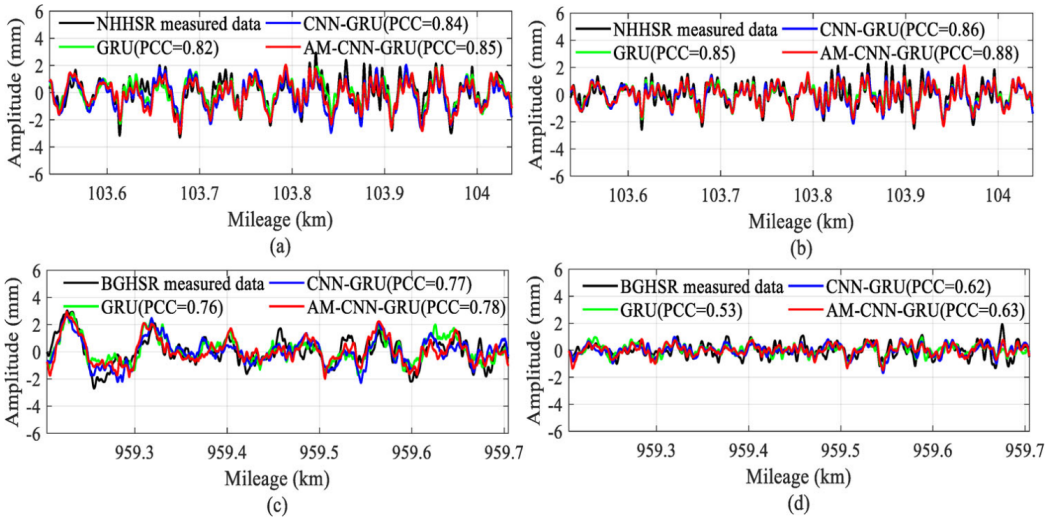


Figure 24 – Reconstruction results comparing the proposed model (red) with two simpler models (blue and green) and the original measurement (black). (***a***) and (***b***) show vertical and lateral, respectively, irregularities reconstruction for the NHHSR line. (***c***) and (***d***) show the vertical and lateral, respectively, irregularities reconstruction for the BGHSR line (HAO et al., 2023).

Although the metrics of Hao et al. (HAO et al., 2023) cannot be directly compared to those of De Rosa et al. (De Rosa; ALFI; BRUNI, 2019) and Tsunashima and Yagura (TSUNASHIMA; YAGURA, 2024), since the former employ different evaluation metrics and the latter do not report any quantitative metrics, the visual comparision show that the proposed approach achieved reconstructions much closer to the real measurements than the model-driven approach and a Gaussian Process Regression model, demonstrating its robustness in handling real-world noisy data. However, the model uses vehicle speed as an input, which may not be optimal. In this thesis, it is proposed instead to use velocity-corrected acceleration values as inputs.

Finally, hybrid methods combine the approaches of model-driven and data-

driven methods (SANSINENA; RODRíGUEZ-ARANA; ARRIZABALAGA, 2025), but the contents of these works are outside the scope of this thesis, since the focus is on data-driven methods.

2.15 Thresholding Acceleration Data

In Rosa et al. (ROSA et al., 2021), the authors try to develop a threshold to detect not safe railway track irregularities. In the paper, the authors propose a machine learning-based classification model that divides the data into two classes based on the standard deviation of the lateral and roll bogie frame accelerations. Class 1 corresponds to normal conditions, i.e., track irregularities below a defined threshold that do not require maintenance, while Class 2 corresponds to sections with irregularities exceeding that threshold, thus requiring maintenance intervention.

The authors created a dataset composed of real and simulated data. The real data were collected using a high-speed TGC, operating at 300 km/h, that was equipped with accelerometers located on the carbody, the bogie frames and the wheelsets. The simulation data was obtained considering a straight track, with a vehicle with speed of 300 km/h and for three different cases of track irregularities: considering only the lateral irregularity, considering only the cross level irregularity, and considering both at the same time. The authors didn't consider vertical irregularities in their simulation. For each case, they ran the simulation 10 times. The authors then applied a band-pass filter in the range of 3 to 27 Hz and then computed the standard deviation of the signals over 100-meter-long track length. Figure 25 shows the whole dataset, where red and blue points are simulation data, black and green are real data, and the red lines shown the standard defined limits for the standard deviation.

Using this dataset, the authors then fitted the data three classifiers: a decision tree, a linear Support vector Machine (SVM) and a Gaussian SVM. Results are shown in Figure 26. The authors, then, discussed the performance of each classifier, but, more importantly for this work, they showed the possibility of defining a threshold in the standard deviation of acceleration data to classify track condition. In the present thesis, it will be tested if this threshold method can also be used in direct acceleration data instead of the standard deviation.

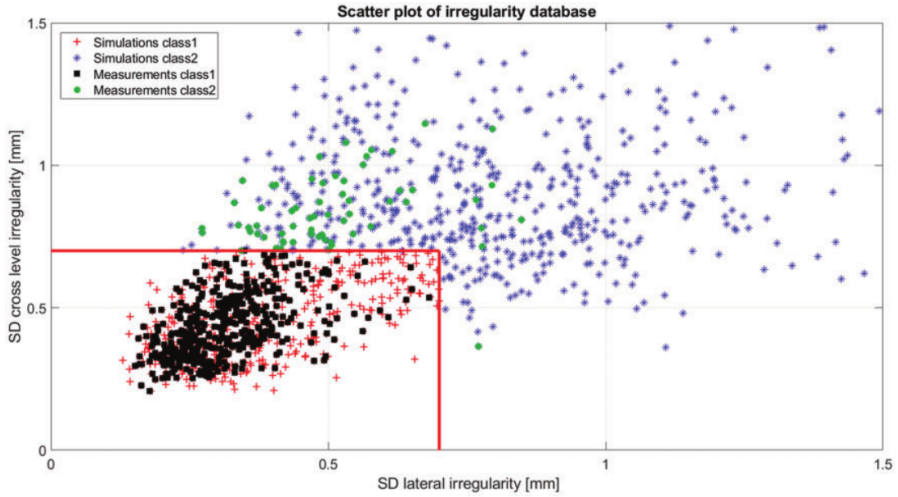


Figure 25 – Dataset distribution of standard deviation of the cross level and lateral track irregularities with defined safety thresholds. Class 1 corresponds to normal conditions, while Class 2 corresponds to sections requiring maintenance intervention (ROSA et al., 2021).

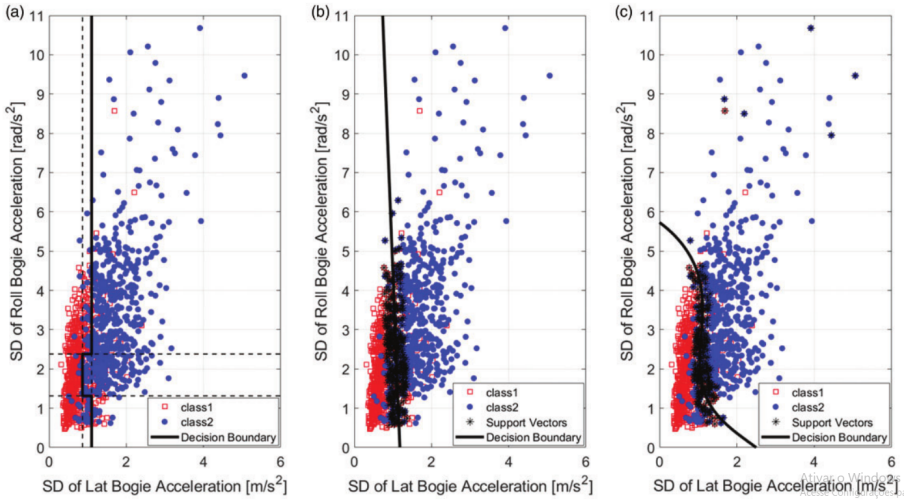


Figure 26 – Classification results of the three classifiers on the test set. (***a***) the decision boundary for the decision tree. (***b***) the decision boundary for the linear SVM. (***c***) the decision boundary for the Gaussian SVM. (ROSA et al., 2021)

3 Methodology

This chapter describes the methodology used in this work. It is organized as follows: Section 3.1 defines the problem to be solved, Section 3.2 describes the simulation parameters for the velocity and mass correction and Section 3.3 describes the machine learning model that will be used to reconstruct the track irregularities.

3.1 Defining the Problem

In this dissertation, the main objective is to develop a unsupervised machine learning model that can reconstruct track irregularities from acceleration data. The reconstructed profiles can be then compared to the standard limits to support maintenance planning, or, in the case of an off limits irregularity, a defect can be identified and repaired before it causes an issue.

Figure 27 shows a flowchart of the step-by-step methodology that will be used to achieve this goal.

To acquire data, a multibody simulation will be set up in the software SIMPACK with the support of the dynamic simulation team, as the implementation details of the model are not the focus of this thesis. The simulation model will first be validated using real data from a section of track geometry measured by the TCG and the corresponding acceleration measurements collected by the IRV. The simulated wagon will replicate the IRV sensor configuration, described in detail in Pires et al. (PIRES et al., 2024), measuring the vertical acceleration on the side frames, the displacement of the secondary suspension, and the triaxial carbody acceleration. Once validated, the simulation will be used to generate a dataset containing sensor data for different track irregularities, velocities, and wagon masses. This search space will be explained in more detail in Section 2.11.

With this dataset, a study will be conducted to analyze the effect of the wagon mass and velocity on the acceleration measurements. The goal is to understand how these parameters affect the data and develop a correction method, using similar approaches described in Sections 2.11 and 2.12, to correct the acceleration data.

The corrected acceleration data will then be used to train a machine learning model for reconstructing track irregularities. Model performance will be evaluated using real-world data. Finally, a threshold will be defined on the reconstructed profiles to determine when maintenance is required, enabling condition-based planning based solely on acceleration measurements.

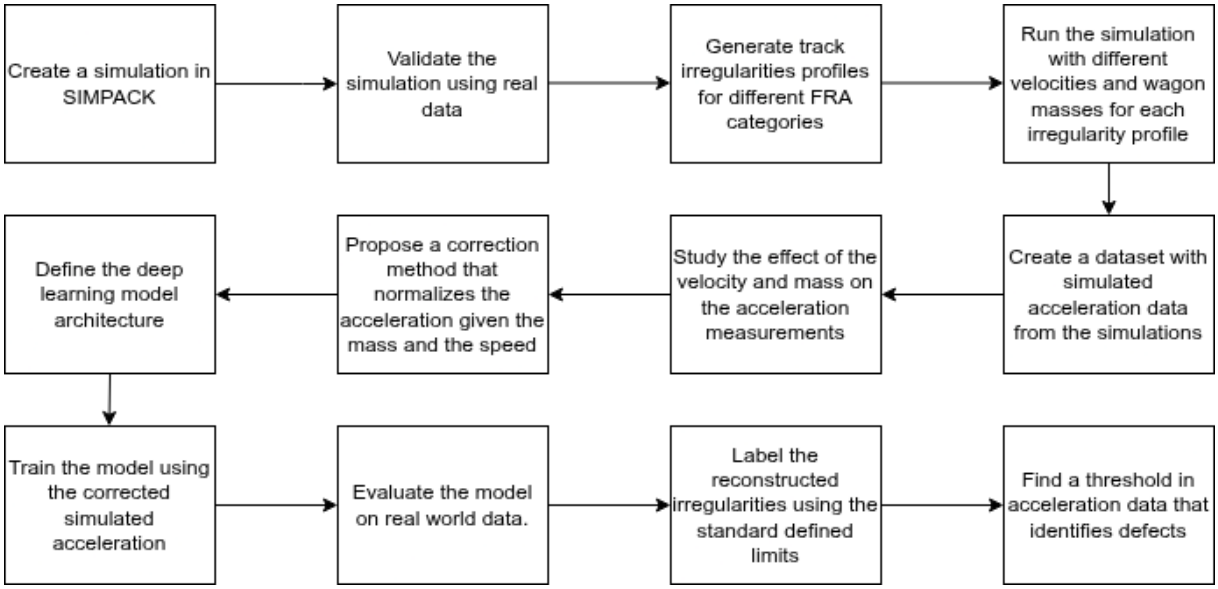


Figure 27 – Flowchart of the proposed methodology.

3.2 Search Space of the SIMPACK Simulation

As stated in Section 3.1 a SIMPACK simulation will run across a search space composed of different velocities, wagon masses and track irregularities. Table 5 summarizes the parameter ranges. Each simulation will test one combination until all possibilities have been explored.

Parameter	**Values**
Velocity (km/h)	5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65
Wagon mass (tons)	0, 51, 102
Track irregularity	No irregularities, FRA 6, FRA 5, FRA 4, Real Irregularities

Table 5 – Search space for the SIMPACK simulation.

The velocity range was defined to cover the typical train velocity operation in the Brazilian Estrada de Ferro Vitória a Minas (EFVM) railway. Wagon masses corresponds to the unloaded wagon class, the half loaded wagon and the fully loaded wagon class. The track irregularities were defined to cover a good track condition (FRA 6), a regular condition track (FRA 5) and a not so good condition track (FRA 4). The "real irregularities" corresponds to a measured track geometry that contains some defects, such as a switch and some misalignments. The "no irregularities" serves as a baseline case to the study.

From this generated data, an approach similar to Balouchi, Bevan and Formston (BALOUCHI; BEVAN; FORMSTON, 2021) will be used to correct the measurements. In this method, a linear or quadratic regression model will be fitted to the absolute maximum acceleration data across different velocities and different wagon masses. The goal is to

develop a correction method that is independent from the track irregularity. The final regression model will be evaluated on real world data.

3.3 Machine Learning Model

The machine learning model to be used in this thesis needs to satisfy a set of requirements. These requirements are:

- The model can not depend on labels, i.e., it needs to be an unsupervised model;
- The model should consider input data as being sequential, i.e., it needs to take into consideration temporal correlations;
- The model needs to be able to handle noisy data;
- The model should be able to be trained on simulated data and perform well on real world data without the need of retraining.

Based on these requirements, the chosen approach is to use a deep learning model that utilizes the methods described in Section 2.4, as it is the most robust approach for handling complex data. These outputs of this model will be then compared to the geometric limits in the Brazilian standard to develop a threshold for maintenance planning based on acceleration data.

4 Schedule

Table 6 shows the schedule of activities to be performed in this thesis from September to December.

	September				October				November				December			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Get and analyze simulation data	⟶															
Find a correction method					⟶											
Define, train and test ML model									⟶				⟶			

Table 6 – Schedule of tasks from September to December

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Appendix

APPENDIX A – Deriving the Backpropagationg Rule

APPENDIX B – Proof of Analytic Signal Theorem

Proof. Assume $z(t)$ is analytic with a real DC component. Analyticity gives $Z(\omega) = S(\omega) + i\,Y(\omega) = 0$ for $\omega < 0$, hence

$$Y(\omega) = i\,S(\omega), \qquad \omega < 0. \tag{B.1}$$

For $\omega > 0$, apply (B.1) at $-\omega < 0$:

$$Y(-\omega) = i\,S(-\omega). \tag{B.2}$$

Taking the complex conjugate of both sides,

$$Y^*(-\omega) = -i\,S^*(-\omega). \tag{B.3}$$

Using Hermitian symmetry $Y^*(-\omega) = Y(\omega)$ and $S^*(-\omega) = S(\omega)$ (valid since $y(t), s(t)$ are real),

$$Y(\omega) = -i\,S(\omega), \qquad \omega > 0. \tag{B.4}$$

Since $y(t)$ is real, $Y(0) \in \mathbb{R}$; likewise $S(0) \in \mathbb{R}$. Hence

$$\mathrm{Im}\{Z(0)\} = \mathrm{Im}\{S(0) + i\,Y(0)\} = Y(0). \tag{B.5}$$

The real-DC hypothesis $\mathrm{Im}\{Z(0)\} = 0$ then forces

$$Y(0) = 0. \tag{B.6}$$

Combining Equations (B.1), (B.4), and (B.6) result in:

$$Y(\omega) = -i\,\mathrm{sgn}(\omega)\,S(\omega) \quad \text{for all } \omega. \tag{B.7}$$

□

APPENDIX C – Real Signals and TFDs

APPENDIX D – Spectral Representation Method

APPENDIX E – An Theoretical Approach for the Dependance of Velocity and Mass

In Sections 2.11 and 2.12, it is shown that the measured accelerations have a dependence with the wagon mass and velocity. In this section, a theoretical approch is presented to explain, in a simplified way, why this dependence exists. Consider, then, a simple model of the wagon represented as a mass-spring-damper system with mass m, spring with stiffness k and damper with damping coefficient c. Assume, also, that the wagon travels with a constant speed v and the track irregularity is given by $r(\chi)$. Figure 28 illustrates this model.

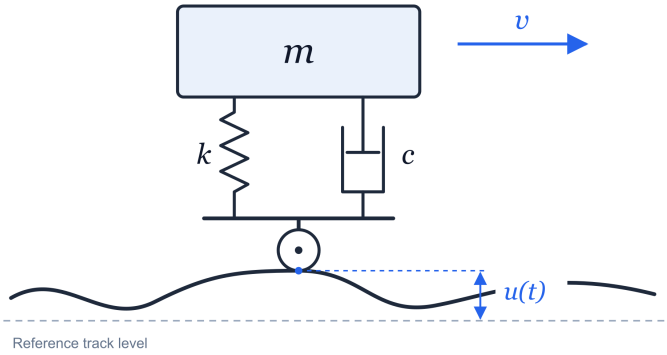


Figure 28 – Mass-spring-damper model of the wagon.

Since the wagon moves at a constant speed v, it is possible to define

$$u(t) = r(\chi(t)) = r(vt) \tag{E.1}$$

where $u(t)$ is the track irregularity as a function of time. Let, then, $z(t)$ be the absolute vertical displacement of the mass, measured from the equilibrium position. The spring force is, then, proportional to the relative displacement $z(t) - u(t)$, and the damping force is proportional to the relative velocity $\dot{z}(t) - \dot{u}(t)$. Using Newton's second law, the equation of motion for the system is:

$$m\ddot{z}(t) = -k(z(t) - u(t)) - c(\dot{z}(t) - \dot{u}(t)) \tag{E.2}$$

which can be rearranged to give

$$m\ddot{z}(t) + c\dot{z}(t) + kz(t) = c\dot{u}(t) + ku(t) \tag{E.3}$$

This is just the equation for the base-excited oscillator (NEWLAND, 2005). Introducing the natural frequency $\omega_n = \sqrt{\frac{k}{m}}$ and the damping ratio $\zeta = \frac{c}{2\sqrt{km}}$, Equation E.3 can be rewritten as

$$\ddot{z(t)} + 2\zeta\omega_n\dot{z}(t) + \omega_n^2 z(t) = 2\zeta\omega_n\dot{u}(t) + \omega_n^2 u(t) \tag{E.4}$$

Since Equation E.3 is a second order ordinary differential equation with constant coefficients, it admits a frequency-domain solution (PEPPIN, 1994). Assume, then, $u(t) = Ue^{i\omega t}$ and $z(t) = Ze^{i\omega t}$. Substituting these expressions into Equation E.4 gives:

$$\left(-m\omega^2 + ic\omega + k\right) Z = \left(ic\omega + k\right) U \tag{E.5}$$

Using this, it is possible to define the displacement transmissibility $H(\omega)$ as:

$$H(\omega) = \frac{Z}{U} = \frac{k + ic\omega}{k - m\omega^2 + ic\omega} \tag{E.6}$$

Rewriting this in terms of ω_n and ζ gives:

$$H(\omega) = \frac{\omega_n^2 + 2i\zeta\omega_n\omega}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \tag{E.7}$$

Defining, then, $x = \frac{\omega}{\omega_n}$, the squared magnitude of the transmissibility can be expressed as:

$$|H(x)|^2 = \frac{1 + 4\zeta^2 x^2}{(1 - x^2)^2 + 4\zeta^2 x^2} \tag{E.8}$$

There are three regimes of interest for the transmissibility. If $x \ll 1$, then $|H(x)|^2 \approx 1$ and the mass follows the track irregularity. If $x \approx 1$, then $|H(x)|^2 \approx \frac{1 + 4\zeta^2}{4\zeta^2} \approx \frac{1}{4\zeta^2}$ for small ζ, and the body is near resonance. Finally, if $x \gg \max(1, 2\zeta)$, then $1 + 4\zeta^2 x^2 \approx 4\zeta^2 x^2$ and $(1 - x^2)^2 + 4\zeta^2 x^2 \approx x^4$, and the transmissibility can be approximated as $|H(x)|^2 \approx \frac{4\zeta^2}{x^2}$. In this last regime, the mass is isolated from the track irregularity.

The above result was derived assuming that the track irregularity profile is a sinusoidal function $u(t) = Ue^{i\omega t}$ with a single frequency ω. However, in reality, the track irregularity is modelled as a realisation of a zero-mean stochastic process, containing a wide range of frequencies (Appendix D):

$$R_r(\xi) = \mathbb{E}[r(\chi)r(\chi + \xi)] = \int_{-\infty}^{\infty} S_r(\omega)e^{i\omega\xi}d\omega \tag{E.9}$$

where $S_r(\omega)$ is the power spectral density of the track irregularity. Using this, the following lemma can be derived:

Lemma E.0.1. *If $r(\chi)$ is a wide-sense stationary in space and $u(t) = r(vt)$ with $v > 0$ constant, then $u(t)$ is wide-sense stationary in time and*

$$S_u(\omega) = \frac{1}{v} S_r\left(\frac{\omega}{v}\right) \tag{E.10}$$

The proof of this lemma is given in Appendix F. From this, it is possible to identify one of the effects of the velocity: the argument $\frac{\omega}{v}$ slides the entire track spectrum to higher frequencies as the velocity increases. This means that, as velocity increases, the excited frequencies of the system also increase.

The solution for the mass-spring-damper system can be written as (NEWLAND, 2005):

$$z(t) = \int_{-\infty}^{\infty} h(\theta)u(t-\theta)d\theta \tag{E.11}$$

where $h(t)$ is a real, absolutely integrable impulse response function. Its autocorrelation function is given by

$$\begin{aligned} R_z(\tau) &= \mathbb{E}\left[\int\int h(\theta_1)h(\theta_2)u(t-\theta_1)u(t+\tau-\theta_2)d\theta_1 d\theta_2\right] \\ &= \int\int h(\theta_1)h(\theta_2)R_u(\tau+\theta_1-\theta_2)d\theta_1 d\theta_2 \end{aligned} \tag{E.12}$$

where the expectation operator was moved inside the integral. Taking the FT and using $\mu = \tau + \theta_1 - \theta_2$ gives:

$$S_z(\omega) = \int\int\int h(\theta_1)h(\theta_2)R_u(\mu)e^{-i\omega(\mu-\theta_1+\theta_2)}d\mu d\theta_1 d\theta_2 \tag{E.13}$$

Since μ, θ_1 and θ_2 are independent variables, the integrals can be separated:

$$\begin{aligned} S_z(\omega) &= \int h(\theta_1)e^{i\omega\theta_1}d\theta_1 \int h(\theta_2)e^{i\omega\theta_2}d\theta_2 \int R_u(\mu)e^{-i\omega\mu}d\mu \\ &= H^*(\omega)H(\omega)S_u(\omega) \end{aligned} \tag{E.14}$$

which gives the result:

$$S_z(\omega) = |H(\omega)|^2 S_u(\omega) \tag{E.15}$$

Now using that in the frequency domain, the acceleration corresponds to multiplication by $(i\omega)^2$, that is $|H_a(\omega)| = \omega^2 H(\omega)$, and combining Equation E.15 and E.10:

$$S_a(\omega) = \frac{\omega^4}{v}|H(\omega)|^2 S_r\left(\frac{\omega}{v}\right) \tag{E.16}$$

Notice that the left-hand side of this equation is a PSD function of the temporal frequency ω, while the right-hand side is a function of the spatial frequency $\frac{\omega}{v}$. These two PSDs are connected by the operator $K_{t\leftarrow\chi}(\Omega; v, m, k, c) = v^3\Omega^4|H(\omega)|^2$. In general, it is desirable to have both PSDs in the χ-domain. Define then $a_\chi(\chi) = a(\frac{\chi}{v})$ the acceleration indexed by distance. Its spatial PSD is given by:

$$S_{a,\chi}(\Omega) = vS_a(v\Omega) \tag{E.17}$$

The Fourier Response Function from $r(\chi)$ to $a_\chi(\chi)$ is then given by:

$$H_{a,\chi}(\Omega) = -(v\Omega)^2 H(v\Omega) \tag{E.18}$$

so that the spatial PSD of the acceleration is given by:

$$S_{a,\chi}(\Omega) = |H_{a,\chi}(\Omega)|^2 S_r(\Omega) = v^4\Omega^4|H(v\Omega)|^2 S_r(\Omega) \tag{E.19}$$

Remember, now, that the variance of a wide-sense stationary process is given by the integral of its PSD over all frequencies, that is:

$$\sigma_a^2 = \int_0^\infty S_a(\omega)d\omega = \int_0^\infty S_{a,\chi}(\Omega)d\Omega = v^4 \int_0^\infty \omega^4|H(\omega)|^2 S_r\left(\Omega\right) d\Omega \tag{E.20}$$

For a specified wavelength band, the PSD can be approximated by a power law, check Section 2.8.1, given by:

$$S_r(\Omega) = A\Omega^{-w}, \quad \Omega_{min} \leqslant \Omega \leqslant \Omega_{max}, \quad 0 < \Omega_{min} < \Omega_{max} < \infty \tag{E.21}$$

For simplicity, assume that the spectrum is approximately zero outside this band. If the spectrum is non-negligible outside this band, their contribution needs to be added to the integral of Equation E.20. Substituting this into Equation E.19 gives:

$$S_{a,\chi}(\Omega) = Av^4\Omega^{4-w}|H(v\Omega)|^2, \quad \Omega_{min} \leqslant \Omega \leqslant \Omega_{max} \tag{E.22}$$

which gives:

$$\sigma_a^2 = Av^4 \int_{v\Omega_{min}}^{v\Omega_{max}} \Omega^{4-w}|H(v\Omega)|^2 d\Omega \tag{E.23}$$

If the contributions of the integral below $v\Omega_{min}$ and above $v\Omega_{max}$ are negligible, then it is possible to approximate the variance as:

$$\sigma_a^2 = Av^4 \int_0^\infty \Omega^{4-w}|H(v\Omega)|^2 d\Omega \overset{\Omega=\omega/v}{=} Av^{w-1} \int_0^\infty \omega^{4-w}|H(\omega)|^2 d\omega = Av^{w-1} I(\omega, \zeta, \omega_n) \tag{E.24}$$

Using this and assuming the vehicle parameters are independent of the velocity:

$$\sigma_a \propto v^{\frac{w-1}{2}} \tag{E.25}$$

This shows that the variance of the acceleration is a power law function of the velocity, with the exponent depending on the value w. In fact, the values of w are limited by the requirement that the integral $I(\omega, \zeta, \omega_n)$ converges. For high frequencies, $|H(\omega)|^2 \approx \dfrac{4\zeta^2\omega_n^2}{\omega^2}$, so the integrand becomes:

$$\omega^{4-w}|H(\omega)|^2 \approx 4\zeta^2\omega_n^2\omega^{2-w} \tag{E.26}$$

which is integrable at infinity if $w > 3$. For low frequencies, $|H(\omega)|^2 \approx 1$, and the integrand becomes:

$$\omega^{4-w}|H(\omega)|^2 \approx \omega^{4-w} \tag{E.27}$$

which is integrable at zero if $w < 5$. Therefore, the exponent w must satisfy $3 < w < 5$.

The influence of the wagon mass, however, is more subtle, as it is embedded within the integral $I(\omega, \zeta, \omega_n)$. It is possible to non-dimensionalize the integral by substituting $\omega = \omega_n x$, which gives:

$$I(\omega, \zeta, \omega_n) = \int_0^\infty (\omega_n x)^{4-w} |H(x)|^2 \omega_n dx = \omega_n^{5-w} \int_0^\infty x^{4-w} J(\zeta, w) dx \tag{E.28}$$

where

$$J(\zeta, w) = \int_0^\infty x^{4-w} \frac{1 + 4\zeta^2 x^2}{(1-x^2)^2 + 4\zeta^2 x^2} dx \tag{E.29}$$

is a pure number that depends only on the damping ratio ζ and the exponent w. Since $\omega_n = \sqrt{\frac{k}{m}}$, $\zeta = \frac{c}{2\sqrt{km}}$, it is possible to write:

$$\sigma_a^2 = A v^{w-1} \left(\frac{k}{m}\right)^{\frac{5-w}{2}} J\left(\frac{c}{2\sqrt{km}}, w\right) \tag{E.30}$$

For the wagon mass, the relationship is more complex, as it depends on the integral $J(\zeta, w)$. Two limiting cases will be presented in Sections E.1 and E.2.

One important observation is that the value of w might change depending on the band $\Omega_{min} \leqslant \Omega \leqslant \Omega_{max}$. Which means that the coefficient of the power law might vary for different frequency bands. For example, in the case of $\Omega \gg \Omega_c$ in Equation 2.50, the PSd behaves as $S_r(\Omega) \propto \Omega^{-4}$, which does not hold for Ω values near Ω_c.

E.1 Lightly Damped System

In the case of the lightly damped system, $\zeta \to 0^+$ the integrand of $J(\zeta, w)$ is dominated by the resonance peak near $x = 1$. In this case, it is possible to approximate $1 - x^2 \approx 2(1-x)$, $1 + 4\zeta^2 x^2 \approx 1$, and $x^{4-w} \approx 1$, so that the integrand becomes:

$$J \approx \int_0^\infty \frac{dx}{4(1-x)^2 + 4\zeta^2} = \frac{1}{4} \int_0^\infty \frac{dx}{(1-x)^2 + \zeta^2} \tag{E.31}$$

Substituting $y = 1 - x$ gives:

$$J \approx \frac{1}{4} \int_{-\infty}^1 \frac{dy}{y^2 + \zeta^2} \approx \frac{1}{4} \int_{-\infty}^\infty \frac{dy}{y^2 + \zeta^2} = \frac{\pi}{4\zeta} \tag{E.32}$$

And the variance of the acceleration becomes:

$$\sigma_a^2 \approx A v^{w-1} \left(\frac{k}{m}\right)^{\frac{5-w}{2}} \frac{\pi\sqrt{km}}{2c} = \frac{\pi A}{2c} v^{w-1} k^{(6-w)/2} m^{(w-4)/2} \tag{E.33}$$

In this case, the dependence of the mass degenerates into a simple power law, with the exponent depending on the value of w:

$$\sigma_a \propto m^{(w-4)/4} \tag{E.34}$$

This dependence is positive if $w > 4$, that is, the acceleration variance increases with mass, and negative if $w < 4$, and the acceleration variance decreases with mass. In the case $w = 4$, the acceleration variance is independent of the mass. One important observation is that the relative error depends on both ζ and w, and is not uniform in the interval $3 < w < 5$.

E.2 Exact Evaluation at $w = 4$

For the case $w = 4$, the integral $J(\zeta, w)$ is:

$$J(\zeta, 4) = \int_0^\infty \frac{1 + 4\zeta^2 x^2}{(1 - x^2)^2 + 4\zeta^2 x^2} dx \tag{E.35}$$

This can be evaluated using the following result:

$$\int_{-infty}^{infty} \left| \frac{B_0 + iB_1 x}{A_0 + iA_1 x - A_2 x^2} \right| dx = \frac{\pi(B_0^2 A_2 + B_1^2 A_0)}{A_0 A_1 A_2} \tag{E.36}$$

Now choosing $A_0 = 1$, $A_1 = 2\zeta$, $A_2 = 1$, $B_0 = 1$, and $B_1 = 2\zeta$, and noticing that the integrand is even, gives:

$$J(\zeta, 4) = \frac{\pi(1 + 4\zeta^2)}{4\zeta} \tag{E.37}$$

Setting $w = 4$ in Equation E.23 gives:

$$\sigma_a^2 = Av^3 \left(\frac{k}{m}\right)^{1/2} \frac{\pi(1 + 4\zeta^2)}{4\zeta} = \frac{\pi A}{2c} v^3 \left(k + \frac{c^2}{m}\right) \tag{E.38}$$

In this case, the variance of the acceleration grows with the third power of the velocity, and decreases with the mass, although this effect is much more subtle, as it is embedded within the term $\frac{c^2}{m}$.

It is important to notice that the results derived in this section are much more qualitative than quantitative, as several approximations were made to simplify the analysis. The real model of a wagon is much more complex than the simple mass-spring-damper system presented here, with, for example, multiple degrees of freedom, nonlinearities, and other effects that were not considered. However, the main goal of this section was to show a more theoretical approach to explain the dependence of the acceleration measurements with the wagon mass and velocity.

APPENDIX F – Proof of the E.0.1 Lemma

Proof. From the definition of the autocorrelation function, it is possible to write:

$$R_u(\tau) = \mathbb{E}[u(t)u(t+\tau)] = \mathbb{E}[r(vt)r(vt+v\tau)] \tag{F.1}$$

Since the mean is constant and the autocorrelation function depends only on τ, u is wide-sense stationary in time. Now, taking the FT and using $\xi = v\tau$:

$$S_u(\omega) = \int_{-\infty}^{\infty} R_r(v\tau)e^{-i\omega\tau}d\tau = \frac{1}{v}\int_{-\infty}^{\infty} R_r(\xi)e^{-i\frac{\omega}{v}\xi}d\xi = \frac{1}{v}S_r\left(\frac{\omega}{v}\right) \tag{F.2}$$

□